

Lecture 17: Bonding in transition metal compounds and coordination complexes-2

Jin Xie (谢进), SPST, ShanghaiTech

Email: xiejin@shanghaitech.edu.cn

Classroom: Teaching Center 401

Wednesday/Friday: 10:15-11:00; 11:10-11:55

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Major points of last lecture

- **Bonding in transition metal compounds and coordination complexes**

1. **Chemistry of the Transition Metals**
2. **Introduction to Coordination Chemistry**
3. **Structures of Coordination Complexes**
4. **Crystal Field Theory: Optical and Magnetic Properties**
5. **Optical Properties and the Spectrochemical Series**
6. **Bonding in Coordination Complexes**

Chemistry of the Transition Metals (过渡金属)

过渡金属元素：元素周期表中d区（ds区包括周期表第IB ~ IIB族元素）。

（不包括镧系和锕系元素，周期表中57~71号元素叫做4f内过渡元素，89~103号元素叫做5f内过渡元素）

Transition metals in the d-block										
Group	3	4	5	6	7	8	9	10	11	12
Period 4	21Sc	22Ti	23V	24Cr	25Mn	26Fe	27Co	28Ni	29Cu	30Zn
5	39Y	40Zr	41Nb	42Mo	43Tc	44Ru	45Rh	46Pd	47Ag	48Cd
6	57La	72Hf	73Ta	74W	75Re	76Os	77Ir	78Pt	79Au	80Hg
7	89Ac	104Rf	105Db	106Sg	107Bh	108Hs	109Mt	110Ds	111Rg	112Cn

Chemistry of the Transition Metals: physical properties

Increasing Z_{eff} from left to right along each period: increasing electron–nuclear attraction

T A B L E 8.1

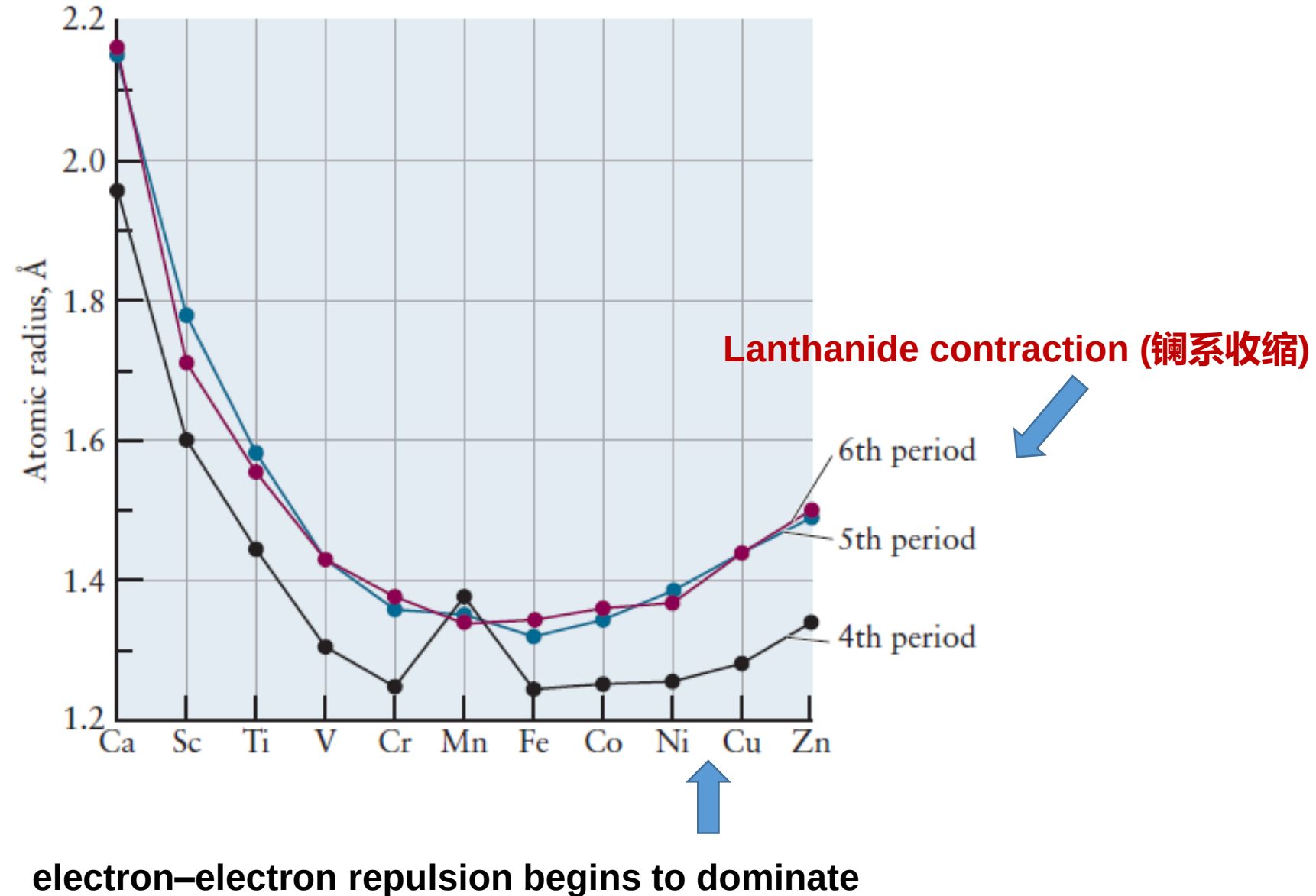
Properties of the Fourth-Period Transition Elements

Element	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
IE_1 (kJ mol ^{−1})	631	658	650	653	717	759	758	737	745	906
IE_2 (kJ mol ^{−1})	1235	1310	1414	1592	1509	1562	1648	1753	1958	1733
Boiling point (°C)	2831	3287	3380	2672	1962	2750	2870	2732	2567	907
Melting point (°C)	1541	1660	1890	1857	1244	1535	1495	1453	1083	420
Atomic radius (Å)	1.61	1.45	1.31	1.25	1.37	1.24	1.25	1.25	1.28	1.34
M ²⁺ ionic radius (Å)	0.81	0.68	0.88	0.89	0.80	0.72	0.72	0.69	0.72	0.74
M ²⁺ configuration	d^1	d^2	d^3	d^4	d^5	d^6	d^7	d^8	d^9	d^{10}
$\Delta H_{\text{hyd}}(\text{M}^{2+})^\dagger$ (kJ mol ^{−1})				−2799	−2740	−2839	−2902	−2985	−2989	−2937

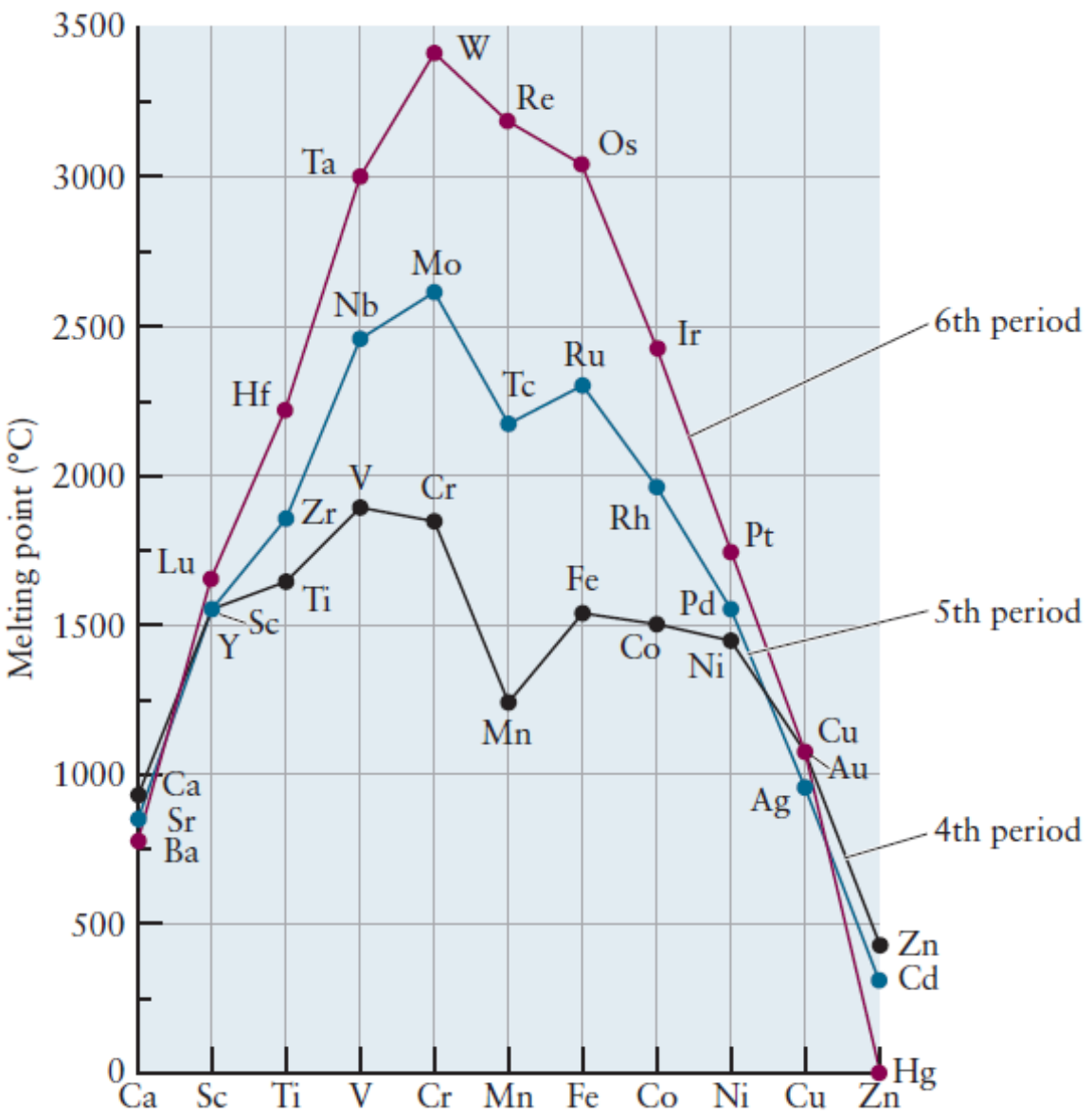
[†]Defined as $\Delta H_f^\circ(\text{M}^{2+}(\text{aq}) - \Delta H_f^\circ(\text{M}^{2+}(\text{g}))$.

Chemistry of the Transition Metals: physical properties

FIGURE 8.1 Variation of atomic radii through the fourth-, fifth-, and sixth-period transition-metal elements. The chemical symbols shown at the bottom of the graph refer to the fourth-period elements.



Chemistry of the Transition Metals: physical properties

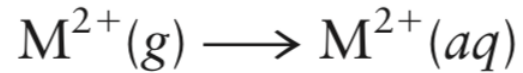


金属键 (metallic bond) 是化学键的一种，
主要在金属中存在。

Metal-metal bond strengths **first increase and then decrease going across each period**, reaching a maximum in the middle.

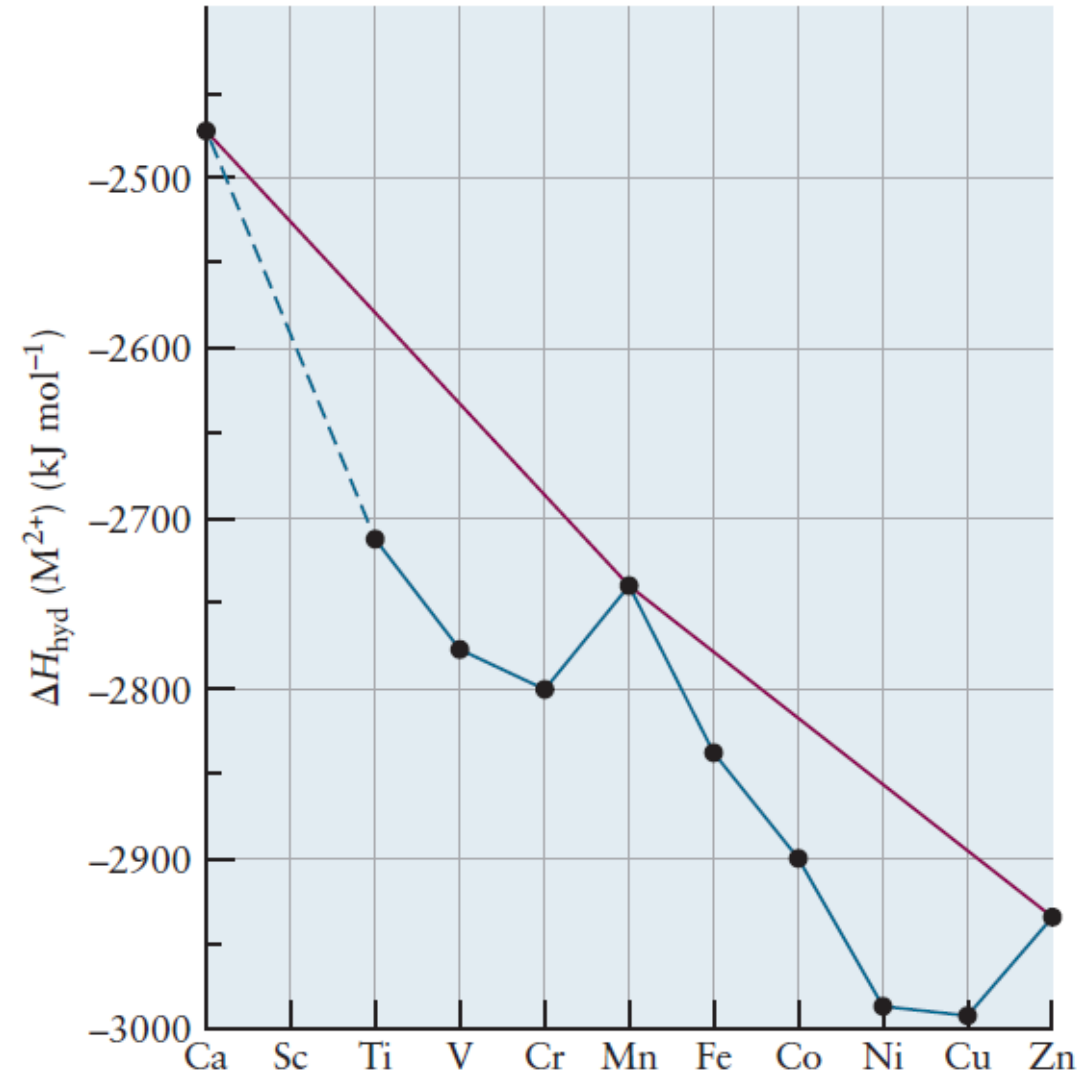
(the number of unpaired electrons in the elements, which reaches a maximum in the middle of the corresponding series.)

Chemistry of the Transition Metals: physical properties



$[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$ ions $[\text{Ni}(\text{NH}_3)_6]^{2+}$ ions

A metal ion is **hydrated (水和)**: the energy change is related to changes in the thermodynamic function **enthalpy (焓)**, denoted by **H**.



strength of the interactions to **increase as the ionic radii decrease**, allowing the water molecules to approach the ions more closely.

Chemistry of the Transition Metals: Oxidation States

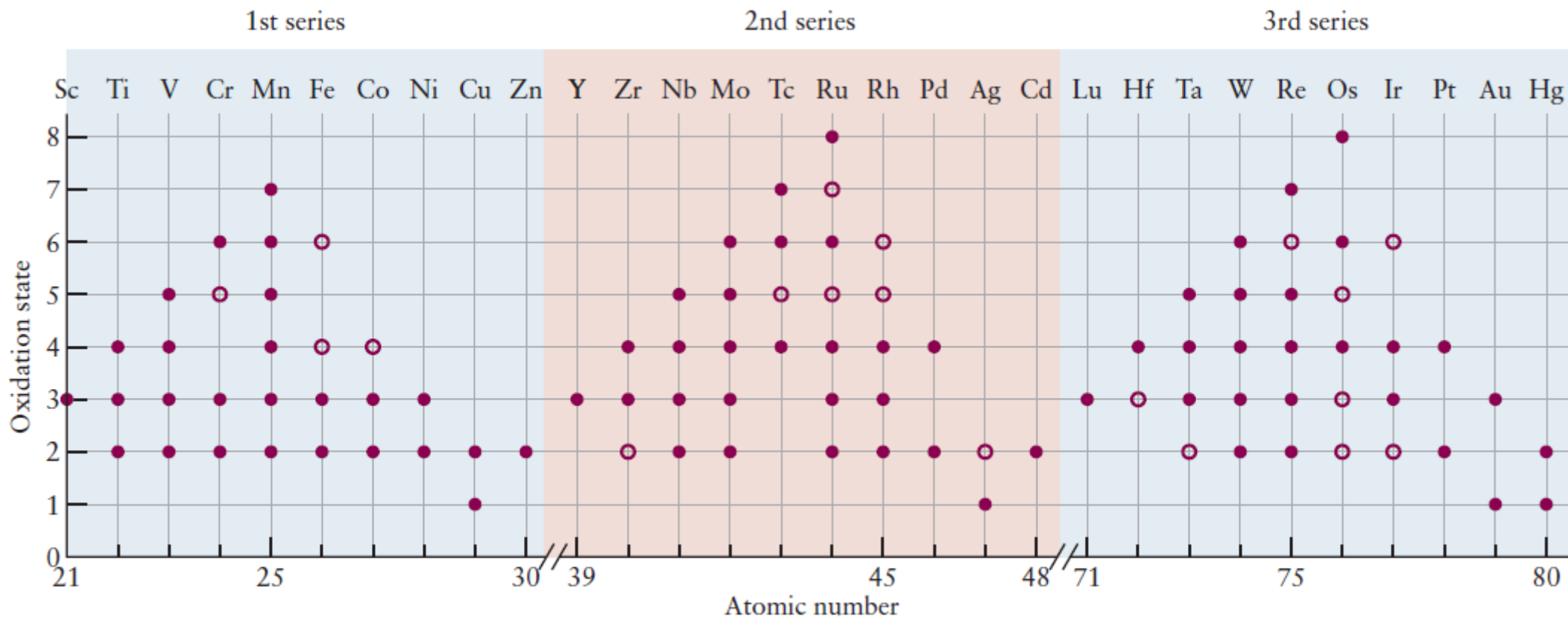


FIGURE 8.4 Some of the oxidation states found in compounds of the transition-metal elements. The more common oxidation states are represented by solid circles, and the less common ones are represented by open circles.

The maximum oxidation states of the **early members** of each period result from the participation of **all the outer s and d electrons in ionic or covalent bonding**.

Chemistry of the Transition Metals: Oxidation States

Compounds with transition metals:

in **high oxidation** states: relatively **covalent**;

in **low oxidation** states: relatively **ionic**.

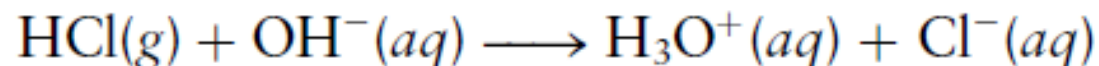
- Mn_2O_7 : covalent compound; it is a liquid at room temperature (crystallizing only at 6°C);
- Mn_3O_4 : ionic compound; containing both Mn(II) and Mn(III), melts at 1564°C .
- A wide range of oxidations states: **partially filled d orbitals** can either **accept or donate electrons** to form chemical bonds.
- Many of their compounds are effective catalysts (催化剂), substances that increase the rates of chemical reactions without undergoing permanent chemical changes themselves.

Chemistry of the Transition Metals: Hard and soft acids and bases

Brønsted-Lowry definition:

Acids (酸): proton donors;

Bases (碱): proton acceptors



Lewis concept:

Acids: electron acceptors;

Bases: electron donors



Hard acid (硬酸): High charge density: small, multiply charged ion

Soft acid (软酸): Low charge density: large, low oxidation state

HSAB

[illegible]

		
hard	intermediate	soft
C	N	O
	P	S
	As	Se
	Sb	Te
		I
		Br
		Cl

“硬亲硬,软亲软,软硬交界都不管”

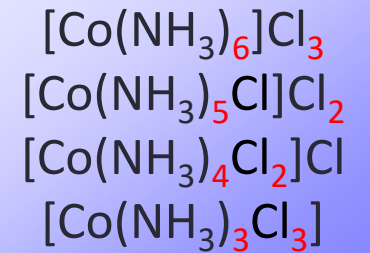
Solubility: $\text{AgF} \gg \text{AgCl} > \text{AgBr} > \text{AgI} \approx \text{Ag}(\text{CN})_2 > \text{Ag}_2\text{S}$

$$\text{CsF} > \text{NaF} \gg \text{LiF} > \text{CaF}_2$$


Ralph G. Pearson
(1919–)

Introduction to Coordination Chemistry

Compound 1: $\text{CoCl}_3 \cdot 6\text{NH}_3$	Orange-yellow	$+ \text{AgNO}_3 \longrightarrow \text{AgCl}$	100%
Compound 2: $\text{CoCl}_3 \cdot 5\text{NH}_3$	Purple		66.7%
Compound 3: $\text{CoCl}_3 \cdot 4\text{NH}_3$	Green		33.3%
Compound 4: $\text{CoCl}_3 \cdot 3\text{NH}_3$	Green		0%



Werner: **two different kinds of species** associated with the cobalt ions

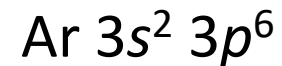
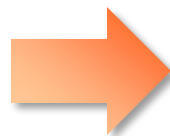
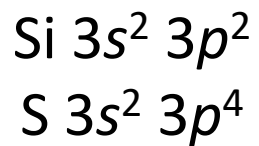
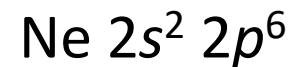
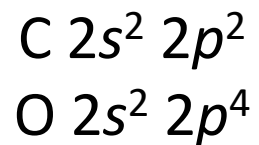
The **primary valences** (主价) were anions like Cl^- in simple salts such as CoCl_3 .

The primary valences were nondirectional. (outer coordination sphere 外界)

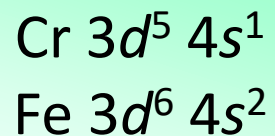
The **secondary valences** (副价) could be either simple anions or neutral molecules such as NH_3 .

The secondary valences were oriented along well-defined directions. (inner coordination sphere 内界)

The 18-Electron Rule (1921)



Octet rule



18-electron rule

Introduction to Coordination Chemistry

Coordination modifies the chemical and physical properties of both the **central atom** and **the ligands**.

CN⁻ ion reacts immediately with acid to generate gaseous HCN, a deadly poison.



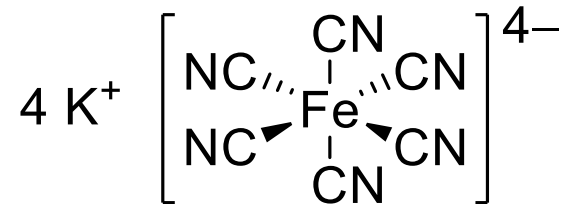
KCN



Fe²⁺, when mixed with aqueous base, instantly precipitates a gelatinous hydroxide.



FeSO₄
· 7H₂O



The reaction between Fe²⁺ and CN⁻ produces the complex ion [Fe(CN)₆]⁴⁻(aq), which undergoes neither of the two reactions just described.

Introduction to Coordination Chemistry

TABLE 8.3

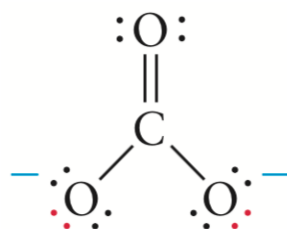
Common Ligands and Their Names

Ligand [†]	Name
:NO ₂ ⁻	Nitro
:OCO ₂ ²⁻	Carbonato
:ONO ⁻	Nitrito
:CN ⁻	Cyano
:SCN ⁻	Thiocyanato
:NCS ⁻	Isothiocyanato
:OH ⁻	Hydroxo
:OH ₂	Aqua
:NH ₃	Ammine
:CO	Carbonyl
:NO ⁺	Nitrosyl

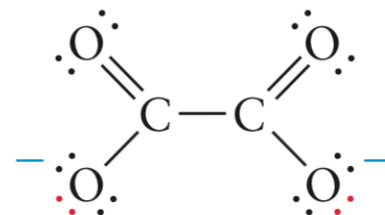
[†]The ligating atom is indicated by a pair of dots (:) to show a lone pair of electrons. In the CO₃²⁻ ligand, either one or both of the O atoms can donate a lone pair to a metal.

Ligands that bond to a metal atom through a single point of attachment are called **monodentate** (单齿) (derived from Latin *mono*, meaning “one,” plus *dens*, meaning “tooth,” indicating that they bind at only one point) **bidentate**, **tridentate**, and so forth.

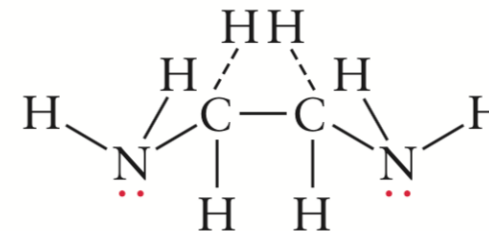
Complexes in which a ligand coordinates via two or more donors to the same central atom are called **chelates** (螯合物)



Carbonate ion,
CO₃²⁻

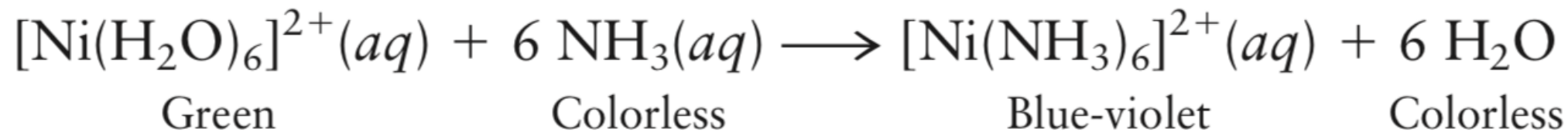
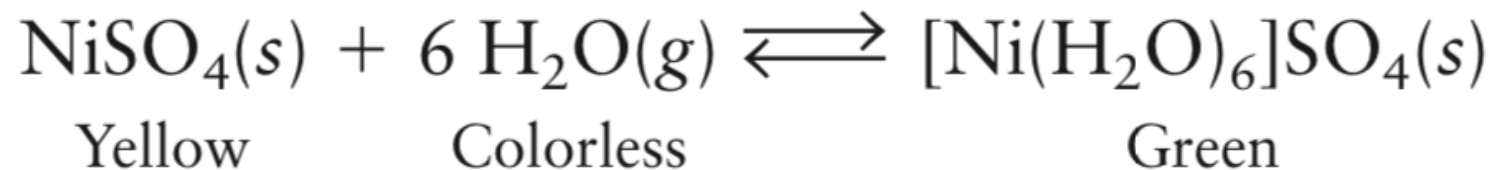


Oxalate ion,
C₂O₄²⁻

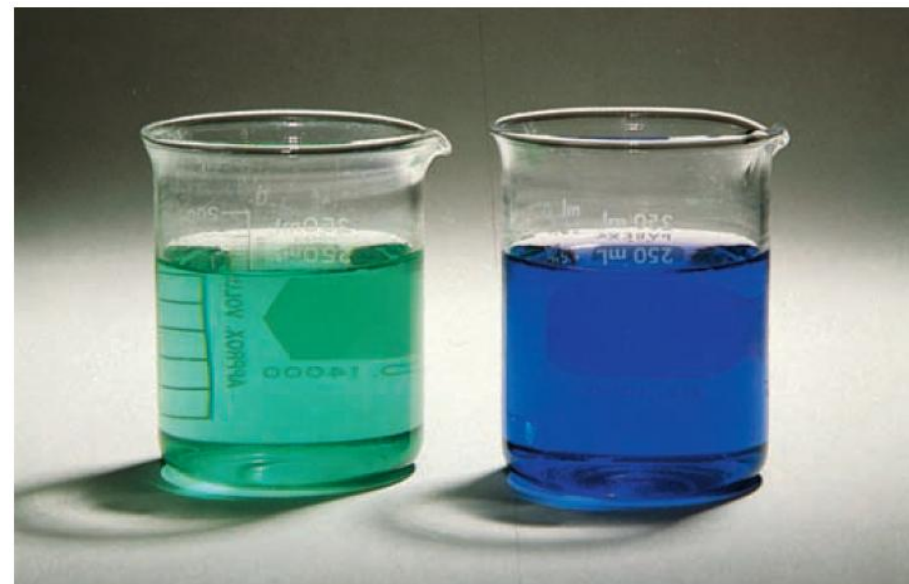


Ethylenediamine,
NH₂CH₂CH₂NH₂

Ligand Substitution (取代) Reactions



Labile (不稳定的) complexes: ligand substitution reactions proceed rapidly. Those in which substitution proceeds slowly or not at all are **inert** (惰性的). These terms are used to describe the different kinetics (rates) of the reactions.



Structures of Coordination Complexes: Octahedral Geometries

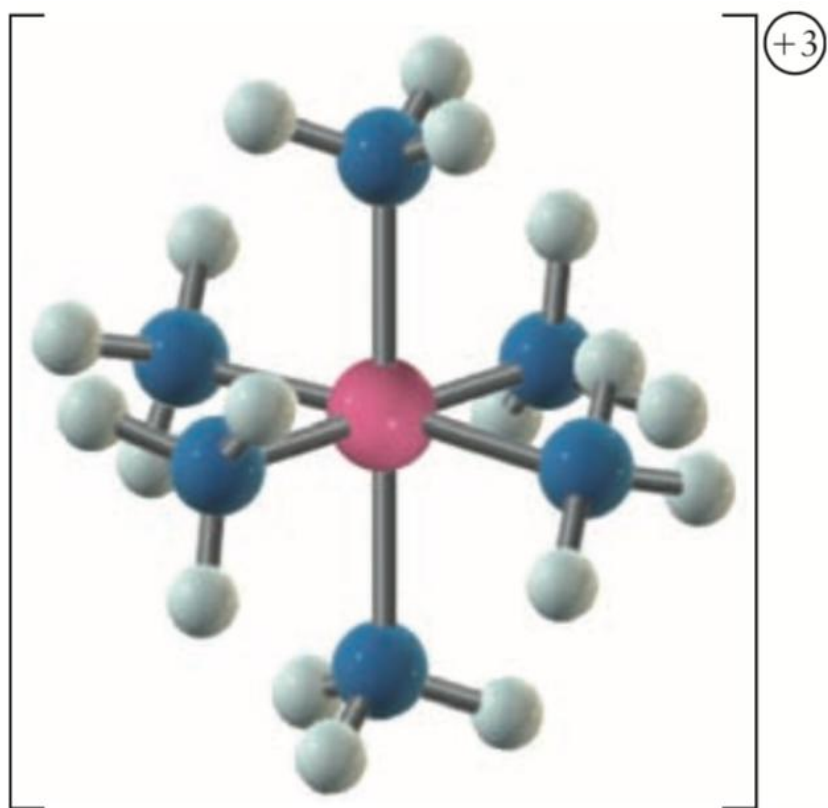
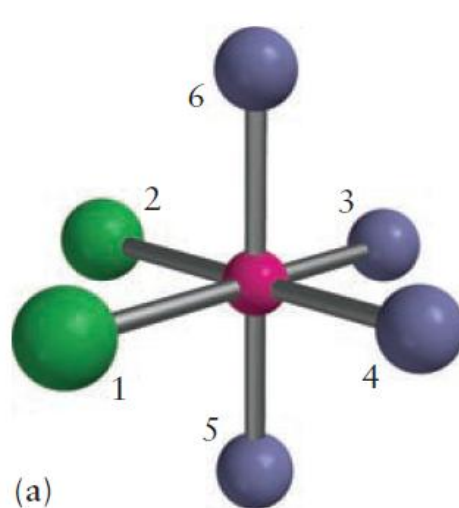
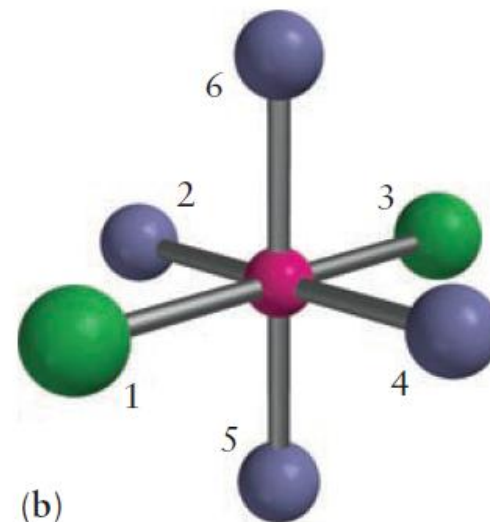


FIGURE 8.14 Octahedral structure of the $\text{Co}(\text{NH}_3)_6^{3+}$ ion. All six corners of the octahedron are equivalent.

$\text{cis-}[\text{Co}(\text{NH}_3)_4\text{Cl}_2]$ $\text{trans-}[\text{Co}(\text{NH}_3)_4\text{Cl}_2]$



(a)

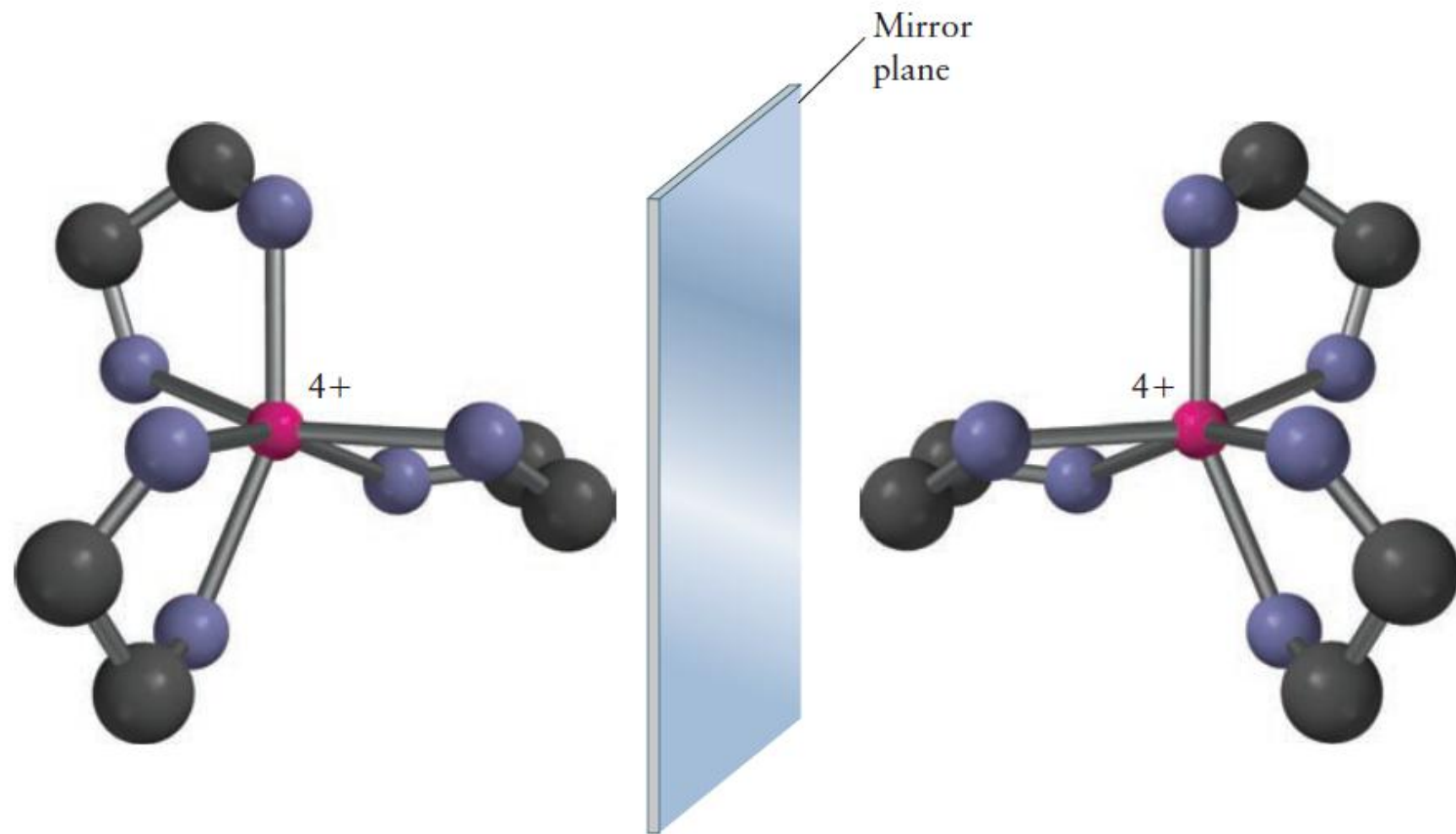


(b)



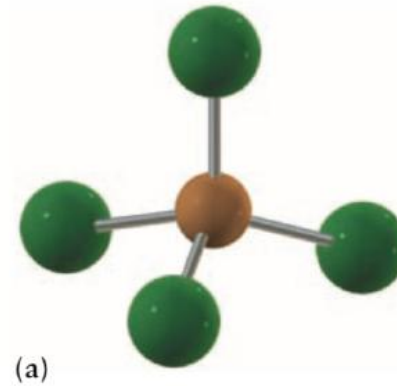
Structures of Coordination Complexes: Chiral Structures

FIGURE 8.13 Enantiomers of the $[\text{Pt}(\text{en})_3]^{4+}$ ion. Reflection through the mirror plane transforms one enantiomer into the other. The two cannot be superimposed by simple rotation.

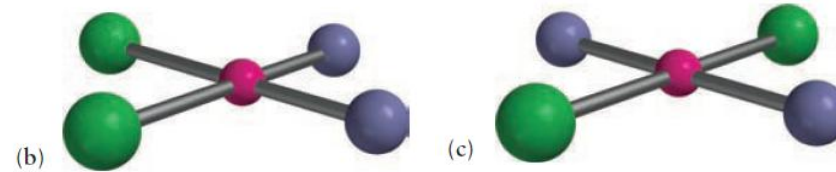


Structures of Coordination Complexes: Square-Planar, Tetrahedral, and Linear

tetrahedral geometry: predominates in the early transition metals (those toward the left side of the d block of elements in the periodic table).



square-planar geometry: is common for four-coordinate complexes of Au^3 , Ir^+ , Rh^+ , and especially common for ions with the d^8 configurations (Ni^{2+} , Pd^{2+} , and Pt^{2+})



(b, c) Square planar, illustrating (b) the *cis* and (c) the *trans* forms of $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$.

linear complexes: coordination numbers of 2, especially common for ions with d^{10} configurations (Cu^{2+} , Ag^+ , Au^+ , and Hg^{2+})

Major points of today's lecture

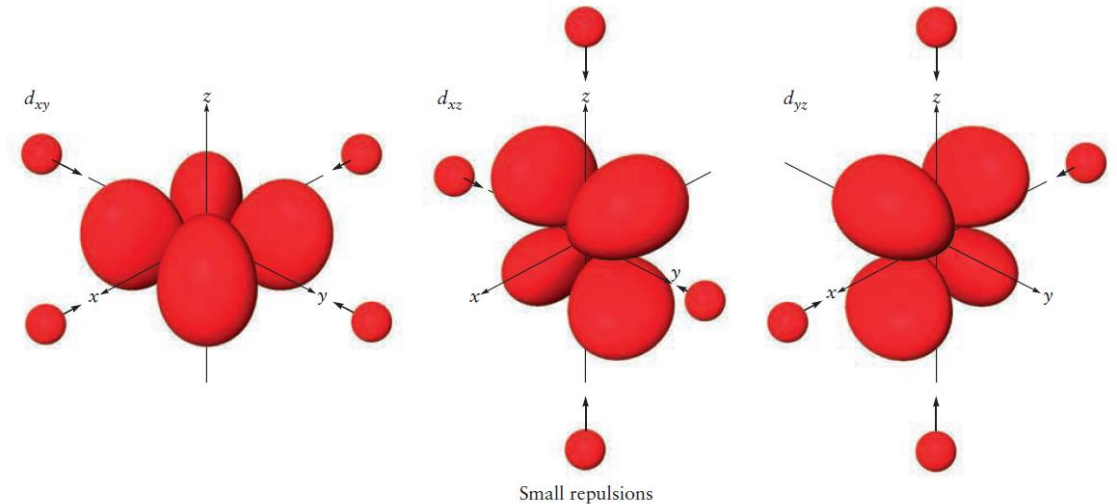
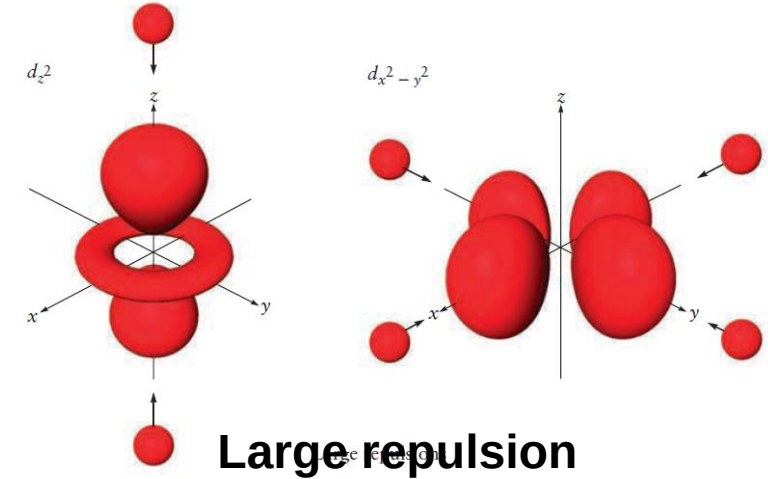
- **Bonding in transition metal compounds and coordination complexes**
 1. **Chemistry of the Transition Metals**
 2. **Introduction to Coordination Chemistry**
 3. **Structures of Coordination Complexes**
 4. **Crystal Field Theory: Optical and Magnetic Properties**
 5. **Optical Properties and the Spectrochemical Series**
 6. **Bonding in Coordination Complexes**

Crystal Field Theory: Optical and Magnetic Properties

Crystal field theory (晶体场理论): an ionic description of metal–ligand bonding.

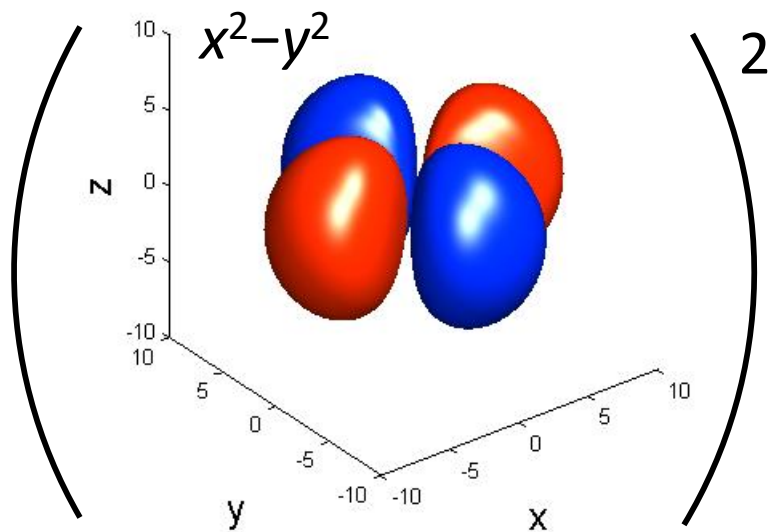
A simple/useful model for understanding the **electronic structure, optical properties, and magnetic properties** of coordination complexes.

Octahedral crystal field

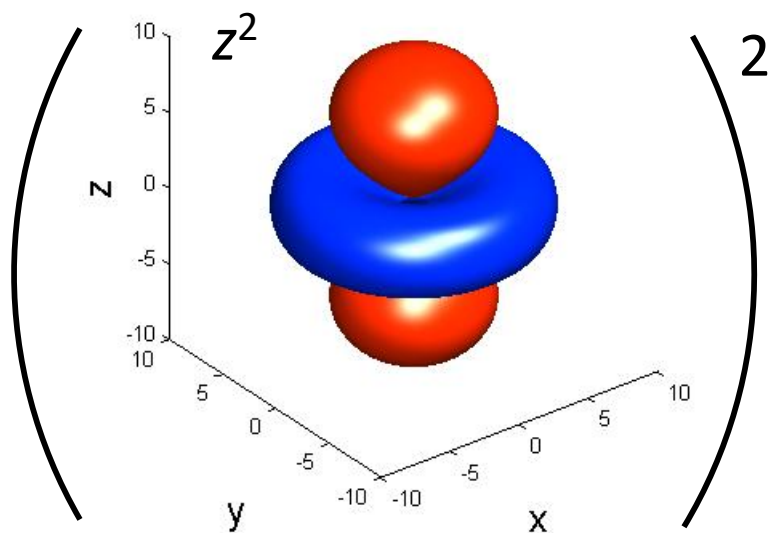


Small repulsion

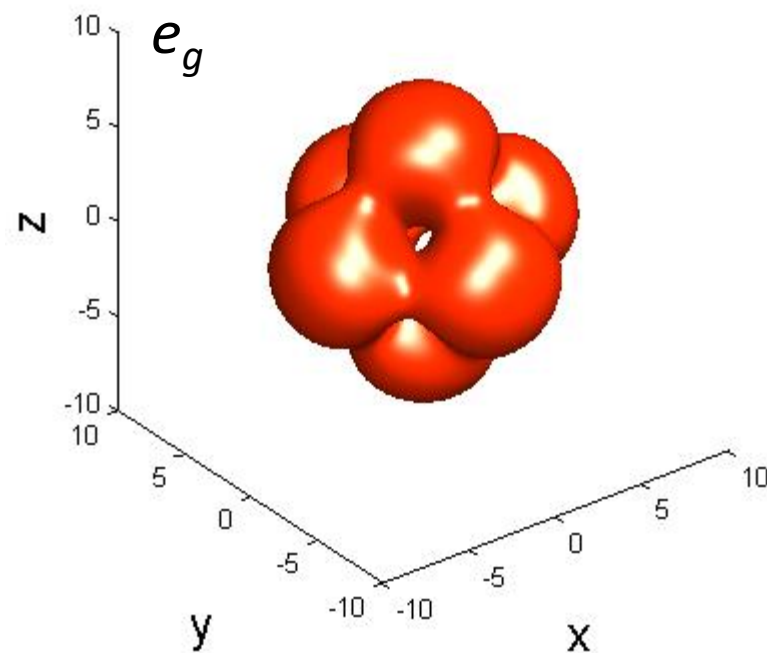
Which *d* Orbitals?



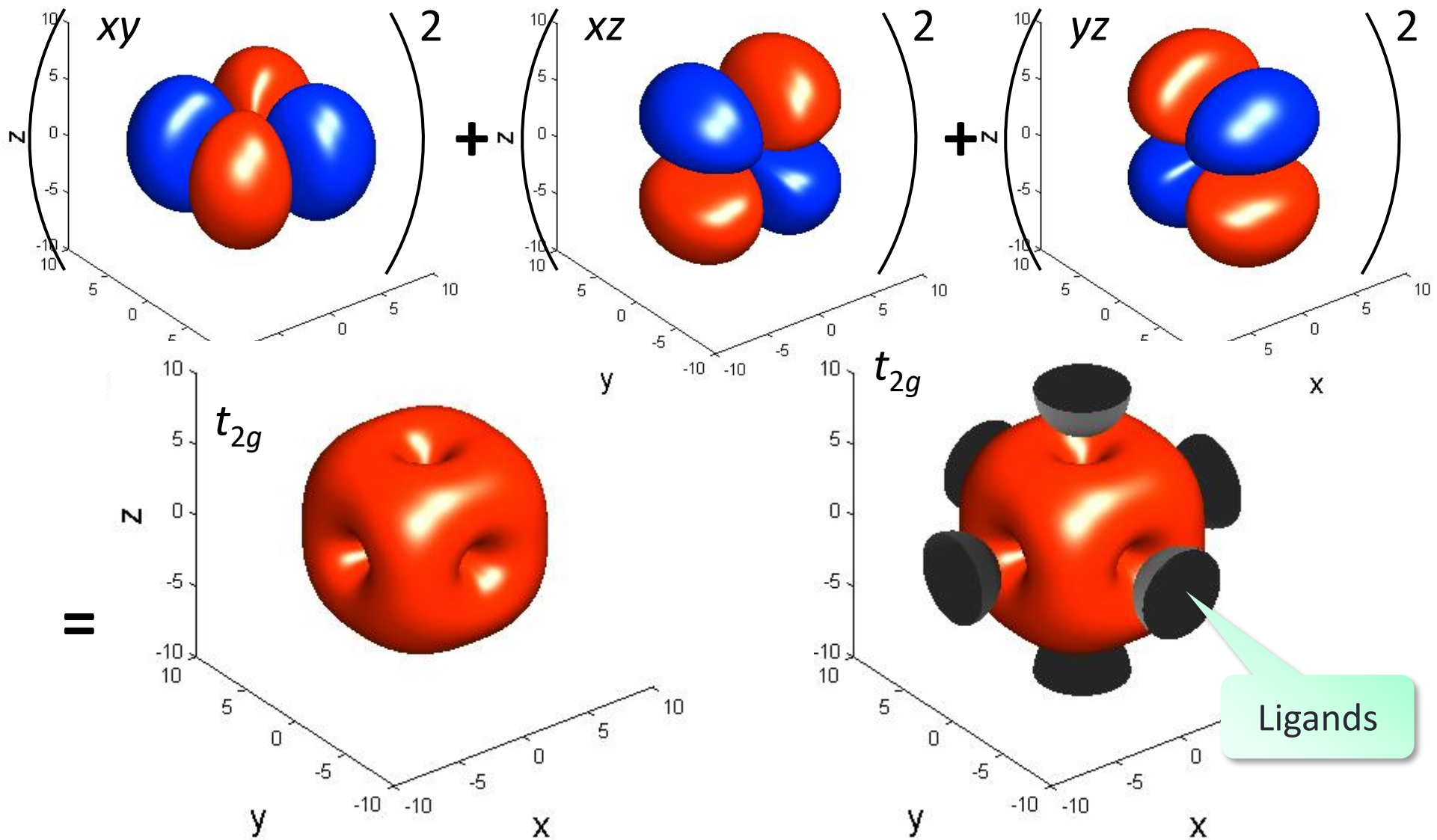
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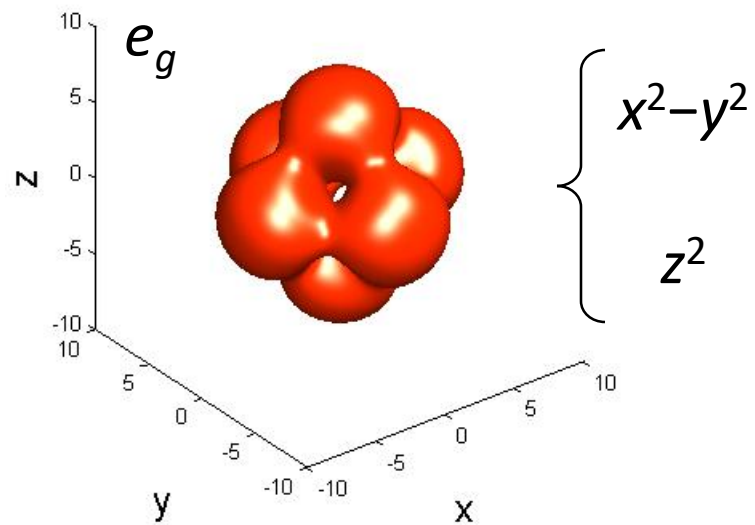
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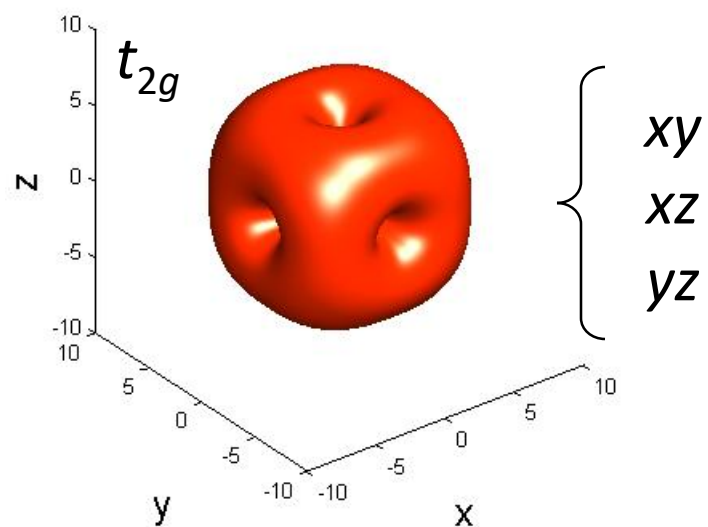
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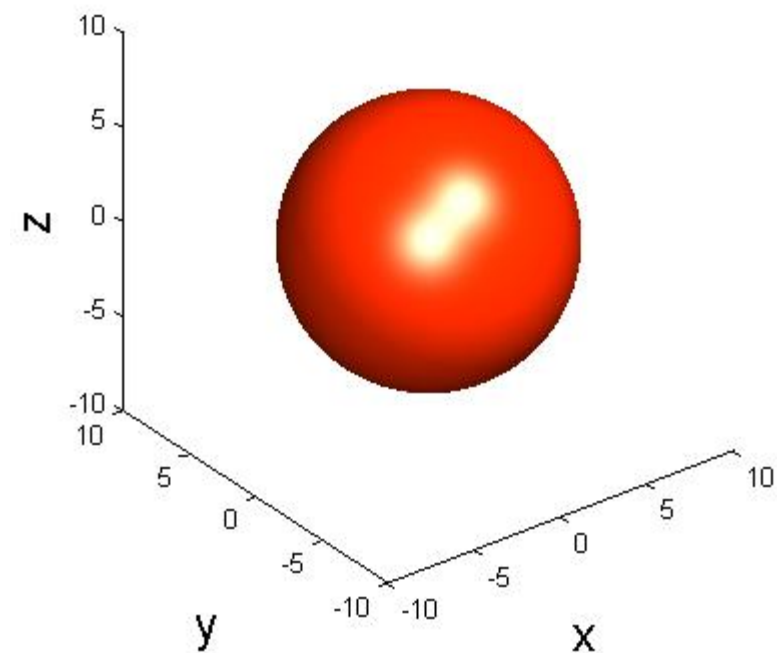
Which *d* Orbitals?



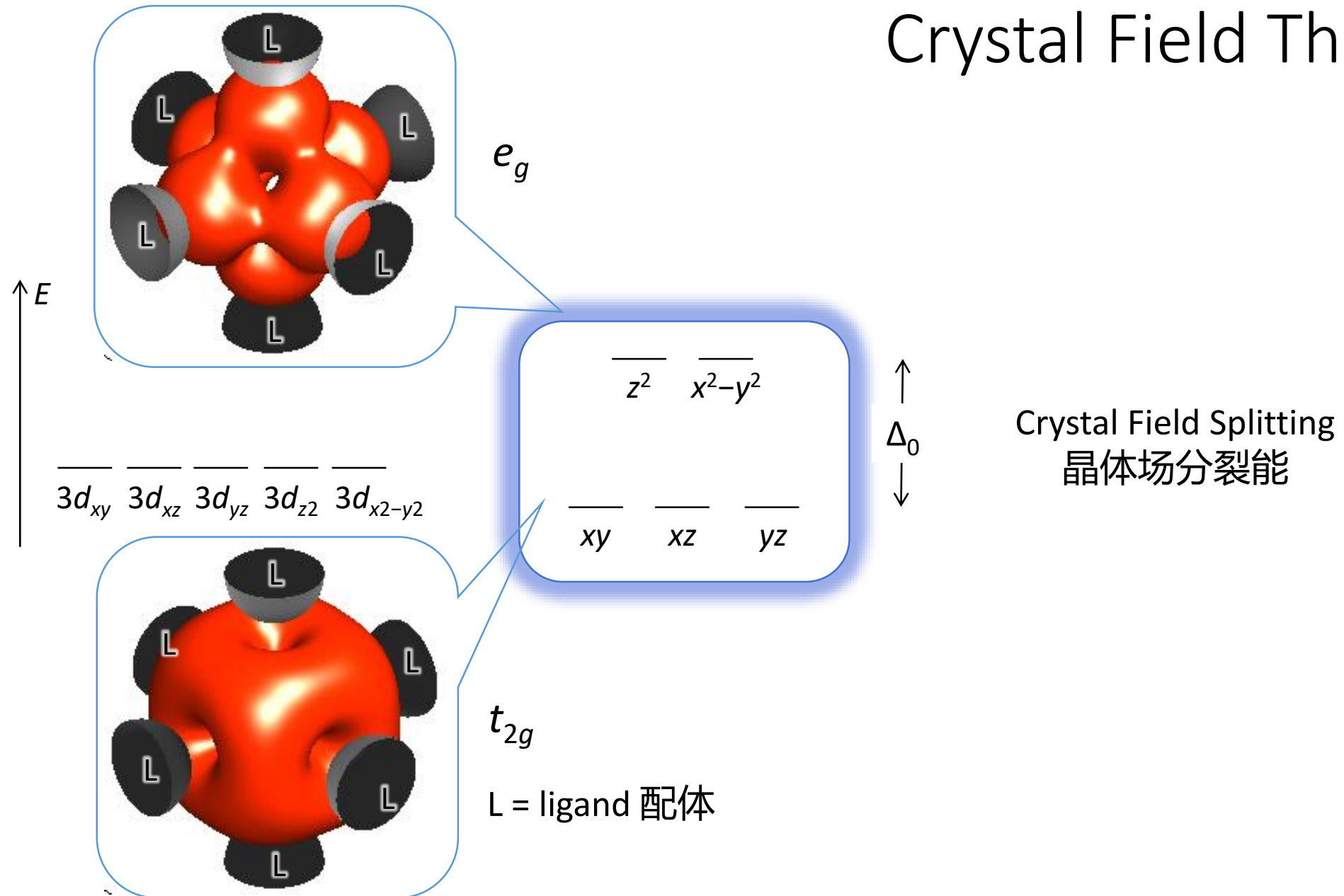
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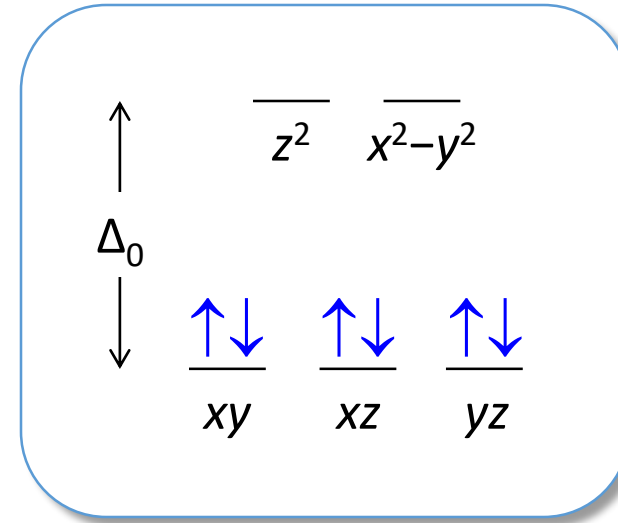
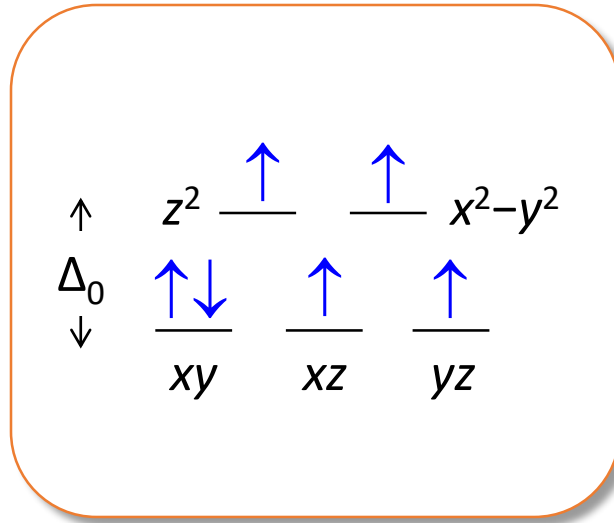
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Crystal Field Theory

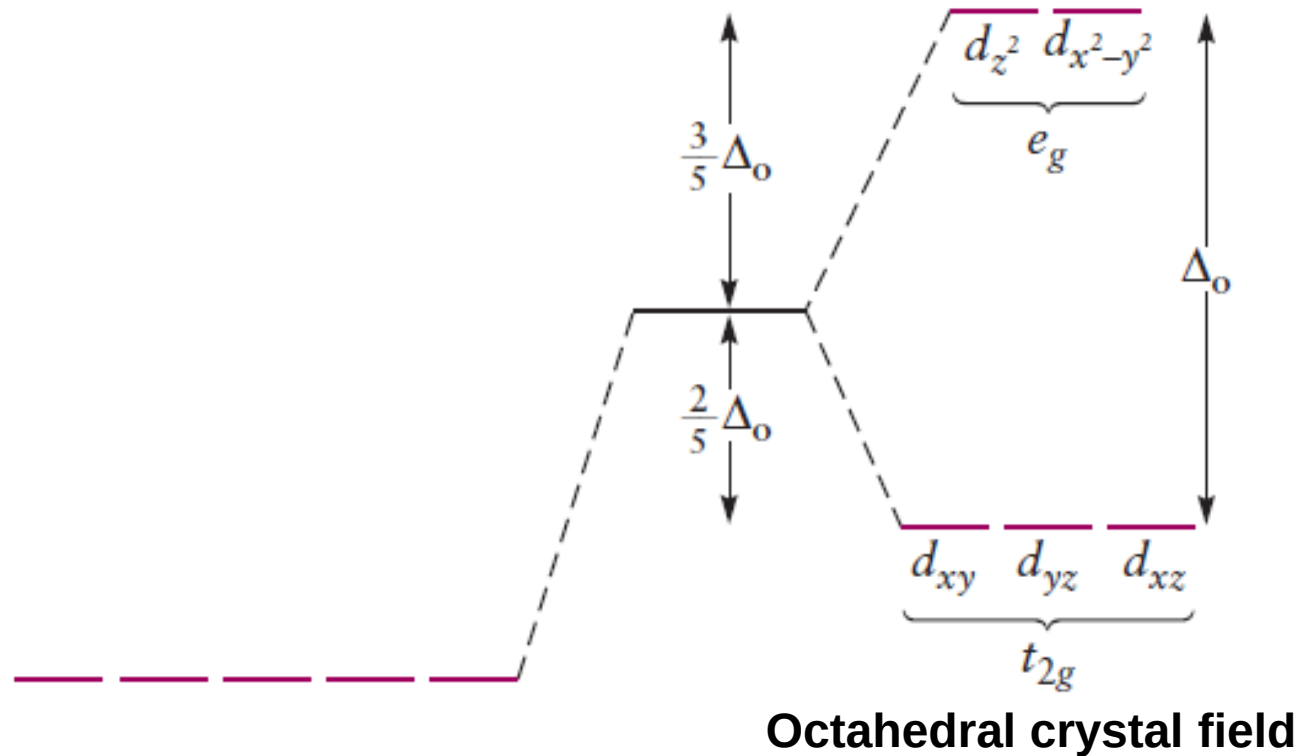


Weak vs. Strong Field Ligands



	CoF_6^{3-}	$\text{Co}(\text{NH}_3)_6^{3+}$
Δ_0	1.80 eV	2.84 eV
Color	Blue, absorbs at 690 nm	Orange, absorbs at 437 nm
18e rule	Does not apply	Applies
Chemistry	Less stable	Stable

Crystal Field Theory: Optical and Magnetic Properties



Δ_o : crystal field splitting energy

In an octahedral field, the energy of the **three** t_{2g} orbitals is lowered by **$\frac{2}{5}\Delta_o$** , and that of the **two** e_g orbitals is raised by **$\frac{3}{5}\Delta_o$** relative to the spherical field.

crystal field stabilization energies (CFSE): energy difference between electrons in an octahedral crystal field and those in a hypothetical spherical crystal field.

Crystal Field Theory: Optical and Magnetic Properties

Octahedral crystal field

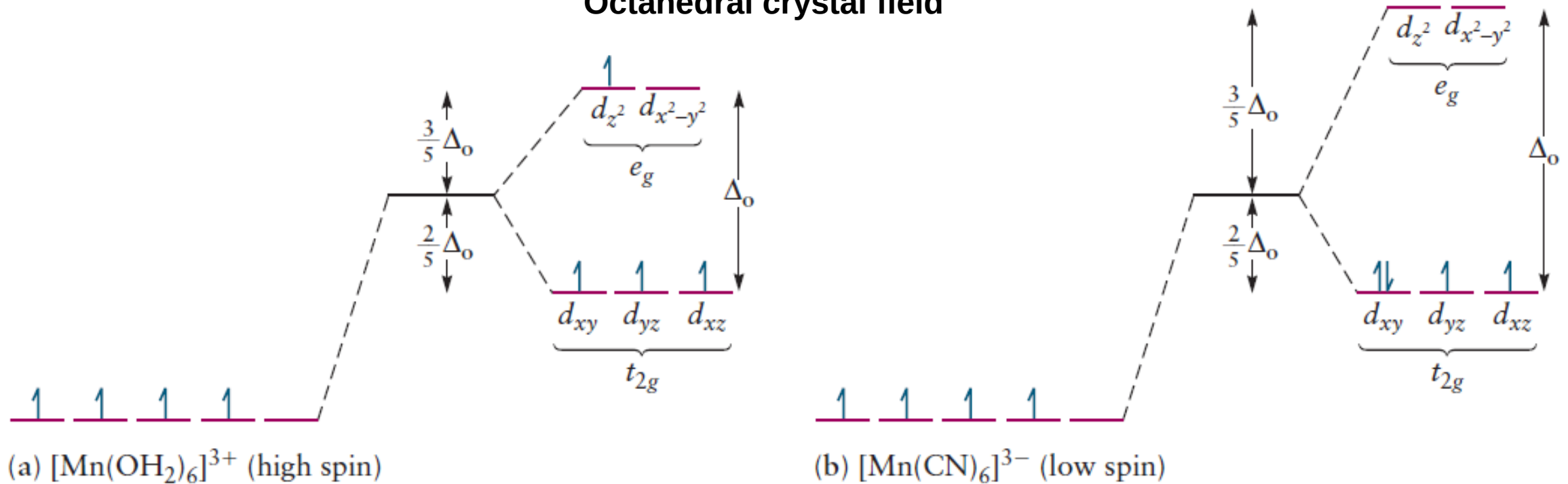


FIGURE 8.18 Electron configuration for (a) high-spin (Δ_o) and (b) low-spin (δ_o) Mn(III) complexes.

Δ_o is small (weak-field ligands):

- high-spin configurations

Δ_o is large (strong-field ligands)

- low-spin configurations

Electrons are paired in the lower energy t_{2g} orbital, and the e_g orbitals are not occupied until the t_{2g} levels are filled.

Crystal Field Theory: Optical and Magnetic Properties

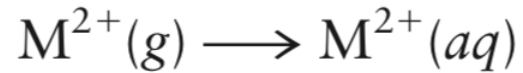
T A B L E 8.5

Electron Configurations and Crystal Field Stabilization Energies for High- and Low-Spin Octahedral Complexes

Configuration		d^1	d^2	d^3	d^4	d^5	d^6	d^7	d^8	d^9	d^{10}
Examples		Ti^{3+}	$\text{Ti}^{2+}, \text{V}^{3+}$	$\text{V}^{2+}, \text{Cr}^{3+}$	$\text{Cr}^{2+}, \text{Mn}^{3+}$	$\text{Mn}^{2+}, \text{Fe}^{3+}$	$\text{Fe}^{2+}, \text{Co}^{3+}$	$\text{Co}^{2+}, \text{Ni}^{3+}$	$\text{Ni}^{2+}, \text{Pt}^{2+}$	Cu^{2+}	Zn^{2+}
HIGH SPIN	e_g	— —	— —	— —	↑ —	↑ ↑	↑ ↑	↑ ↑	↑ ↑	↑ ↓ ↑	↑ ↓ ↑ ↓
	t_{2g}	↑ — —	↑ ↑ —	↑ ↑ ↑	↑ ↑ ↑	↑ ↑ ↑	↑ ↓ ↑ ↑	↑ ↓ ↑ ↓ ↑	↑ ↓ ↑ ↓ ↑ ↓	↑ ↓ ↑ ↓ ↑ ↓	↑ ↓ ↑ ↓ ↑ ↓
	CFSE	$-\frac{2}{5} \Delta_o$	$-\frac{4}{5} \Delta_o$	$-\frac{6}{5} \Delta_o$	$-\frac{3}{5} \Delta_o$	0	$-\frac{2}{5} \Delta_o$	$-\frac{4}{5} \Delta_o$	$-\frac{6}{5} \Delta_o$	$-\frac{3}{5} \Delta_o$	0
LOW SPIN	e_g				— —	— —	— —	↑ —			
	t_{2g}				↑ ↓ ↑ ↑	↑ ↓ ↑ ↓ ↑	↑ ↓ ↑ ↓ ↑ ↓	↑ ↓ ↑ ↓ ↑ ↓			
	CFSE	Same as high spin			$-\frac{8}{5} \Delta_o$	$-\frac{10}{5} \Delta_o$	$-\frac{12}{5} \Delta_o$	$-\frac{9}{5} \Delta_o$	Same as high spin		

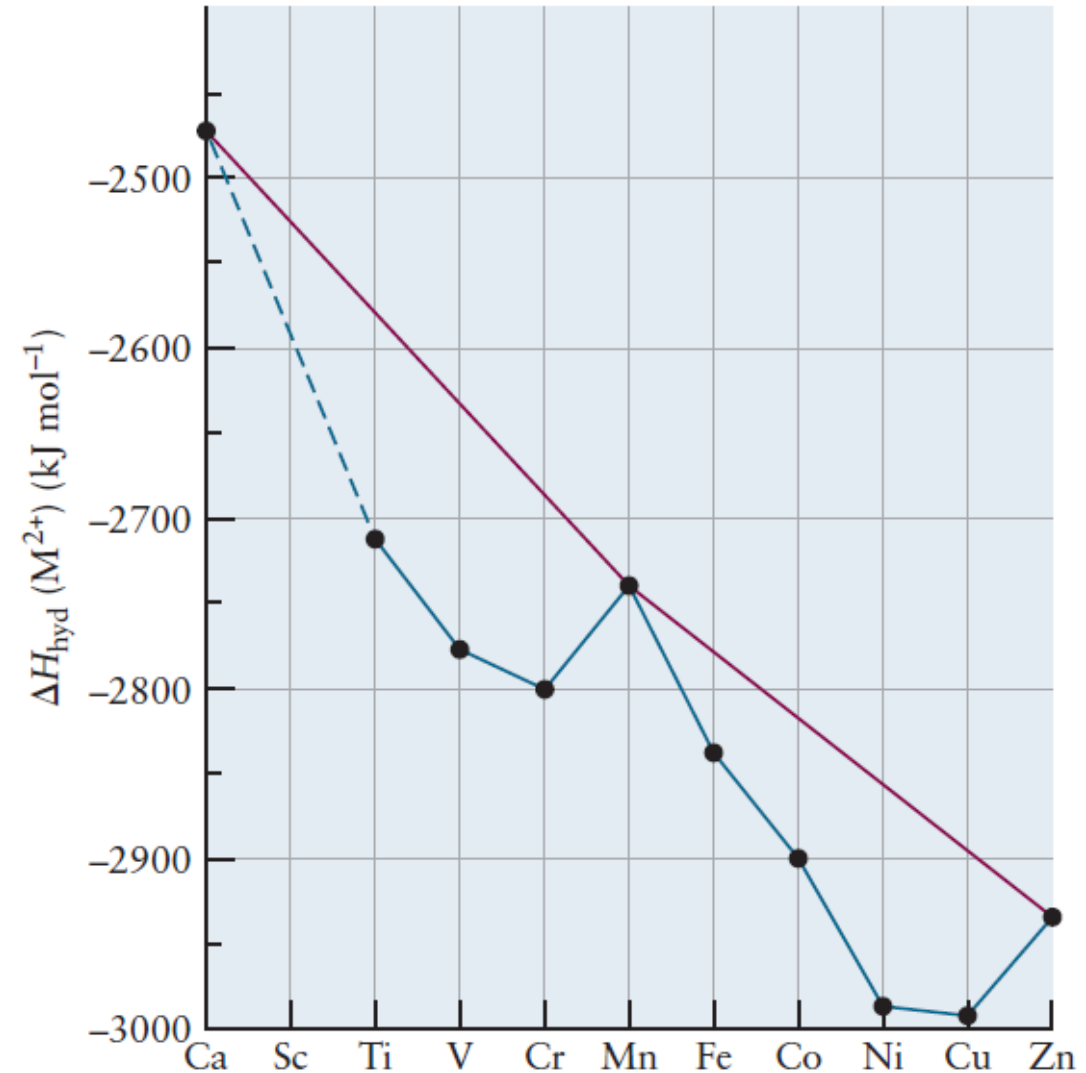
CFSE, Crystal field stabilization energies.

Chemistry of the Transition Metals: physical properties



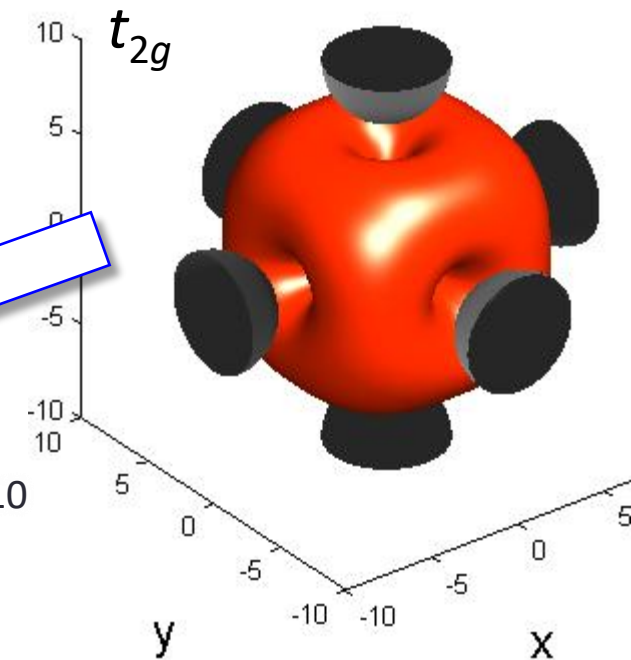
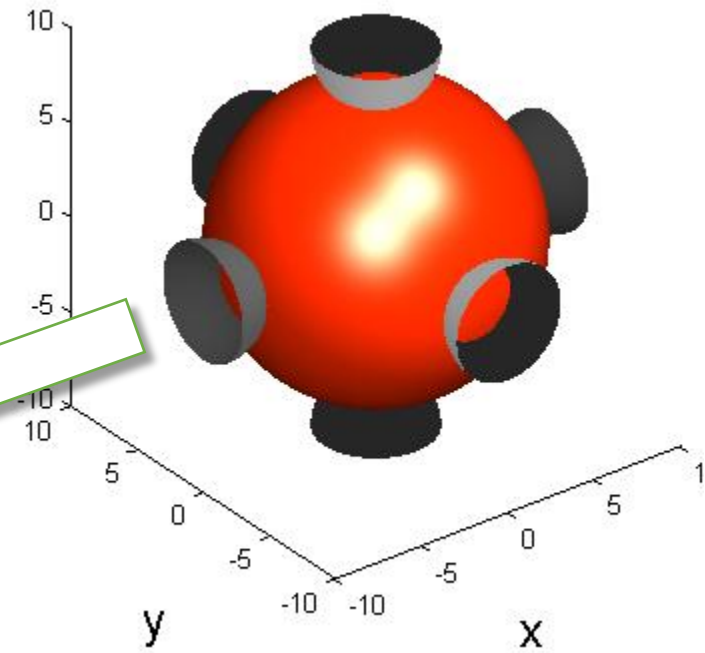
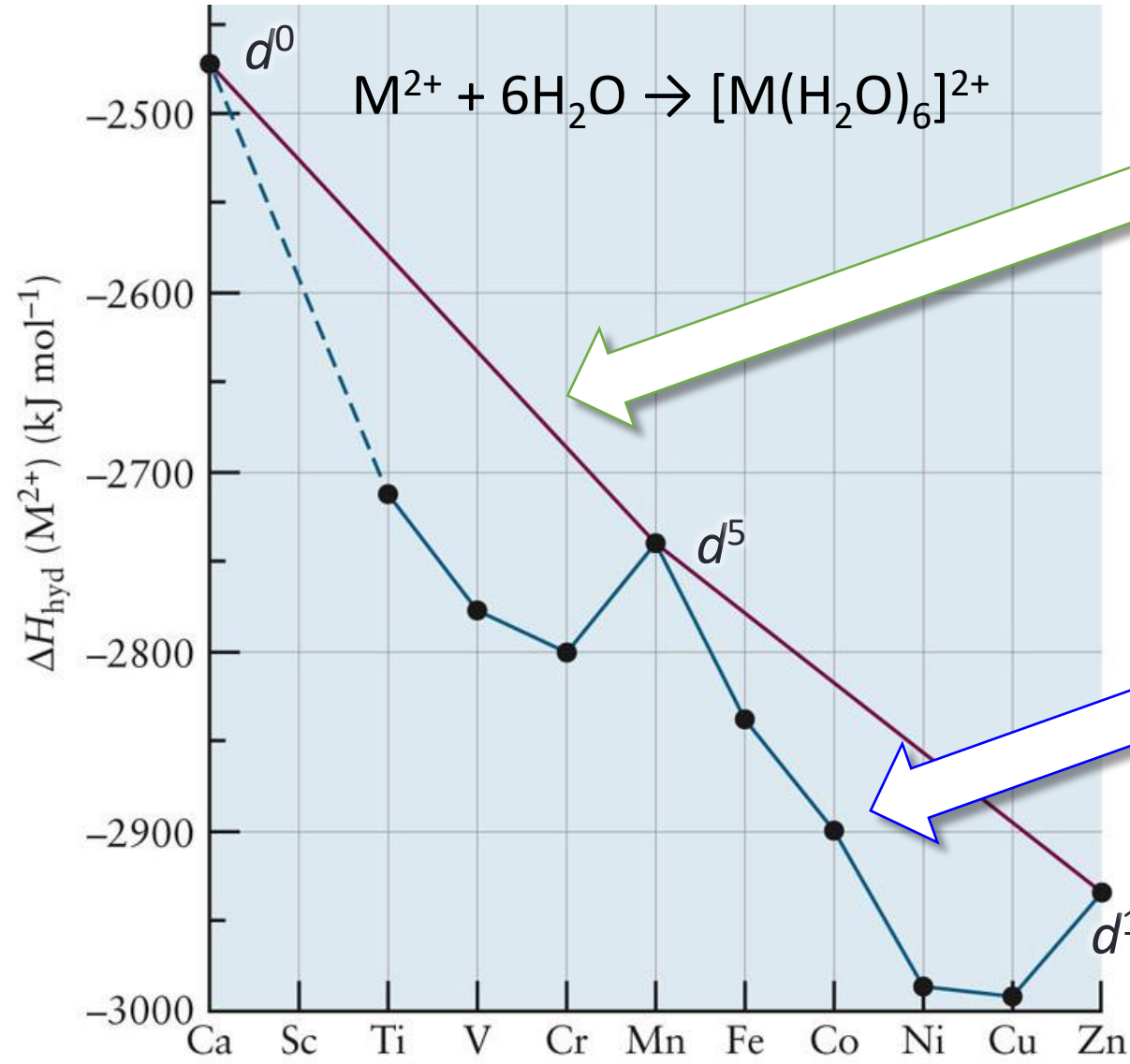
$[Ni(H_2O)_6]^{2+}$ ions $[Ni(NH_3)_6]^{2+}$ ions

A metal ion is **hydrated (水和)**: the energy change is related to changes in the thermodynamic function **enthalpy (焓)**, denoted by **H**.

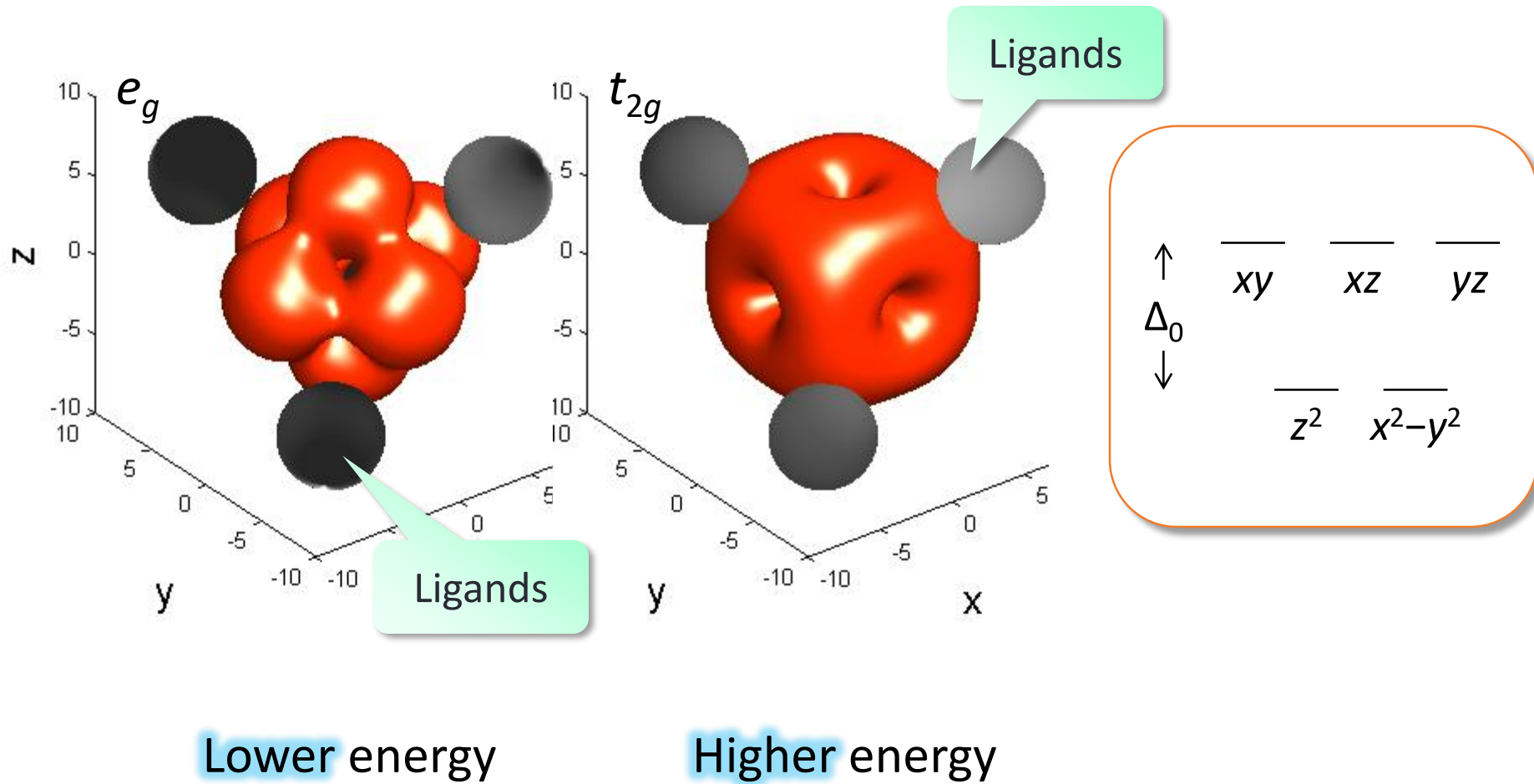


strength of the interactions to **increase as the ionic radii decrease**, allowing the water molecules to approach the ions more closely.

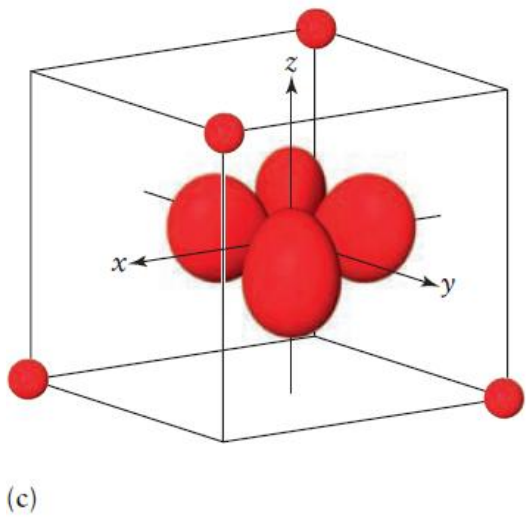
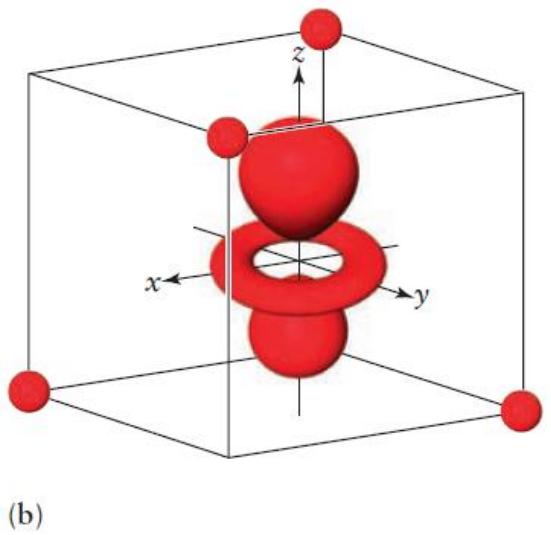
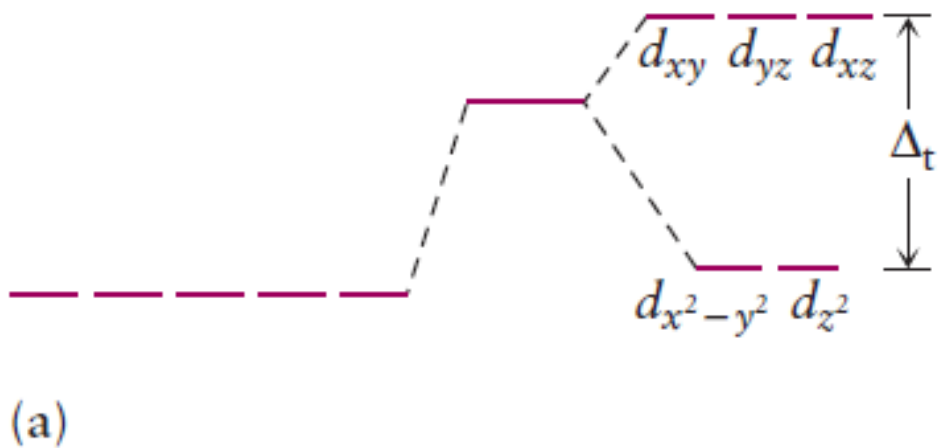
Heat of Hydration



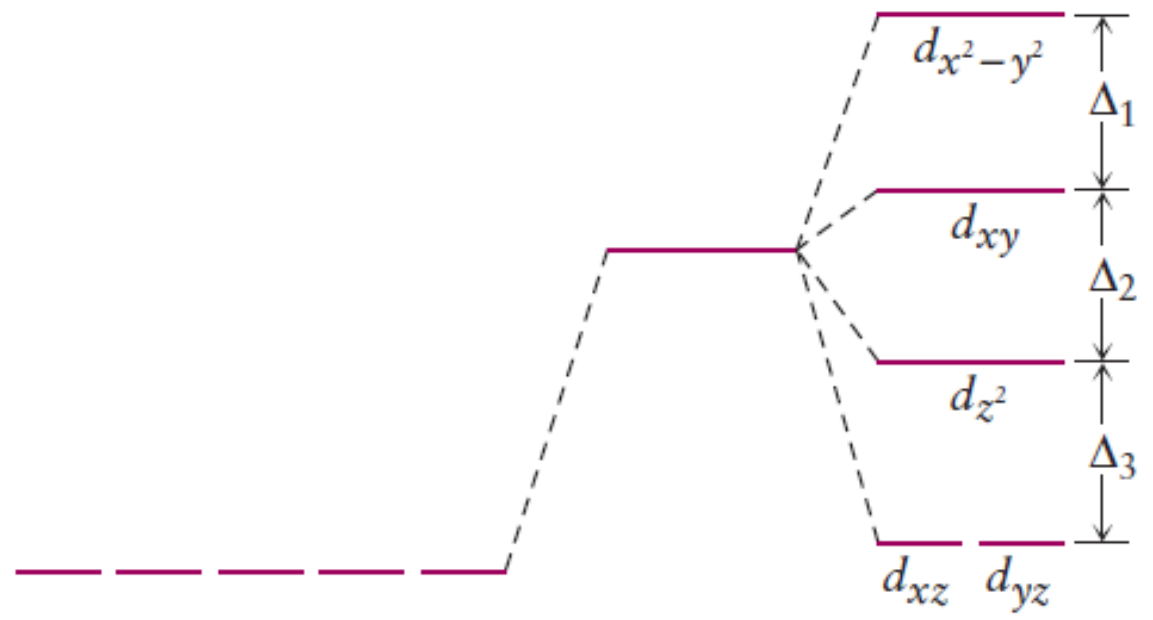
Tetrahedral Crystal Field



Crystal Field Theory: square-planar and tetrahedral complexes

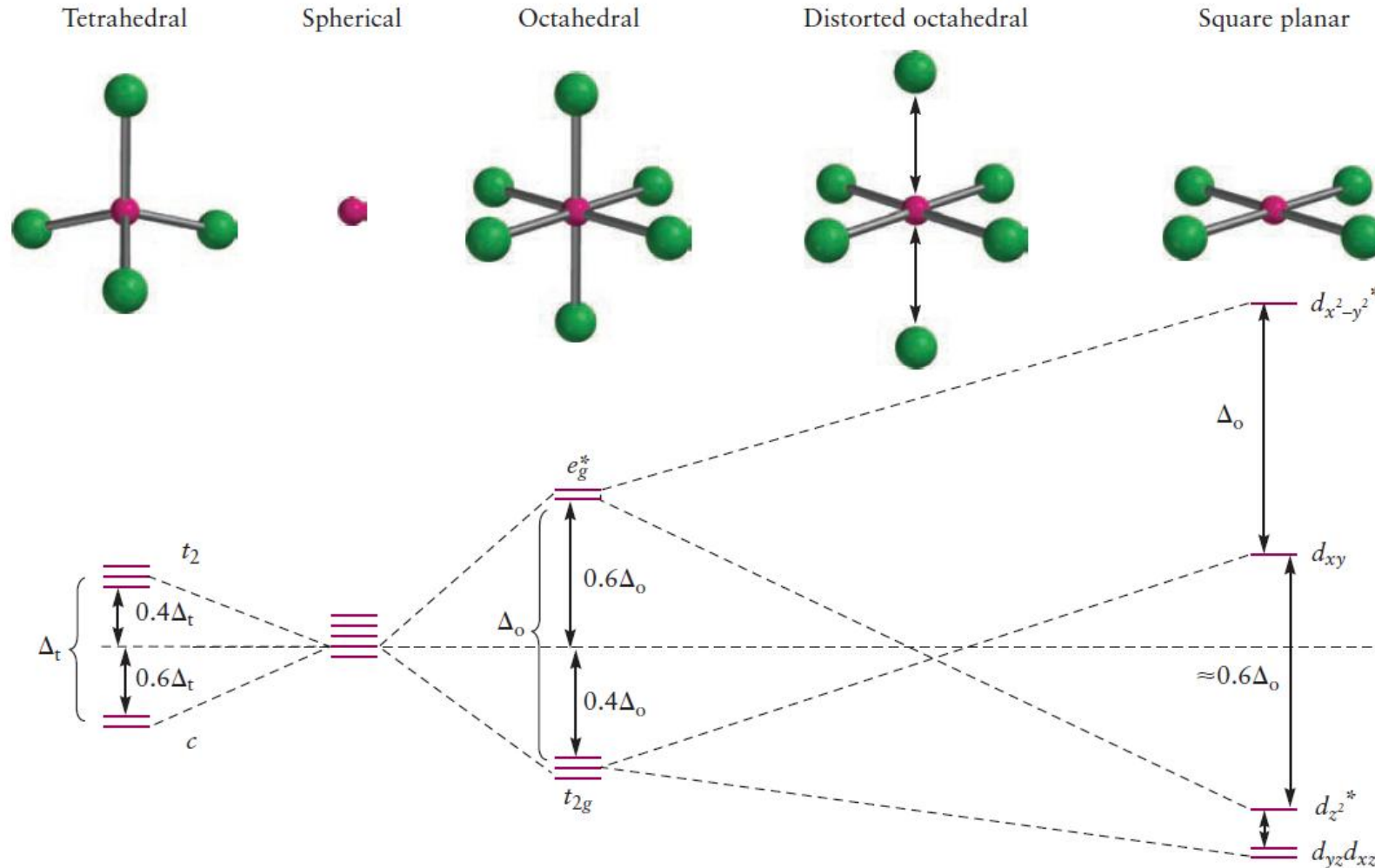


A tetrahedral crystal field



A square planar crystal field

Crystal Field Theory: summary



Tetrahedral: less stable

- The crystal field splitting is smaller
- The lower-energy set of levels is only doubly degenerate

Octahedral: most common:

- the formation of 6 bonds, rather than 4, confers greater stability.

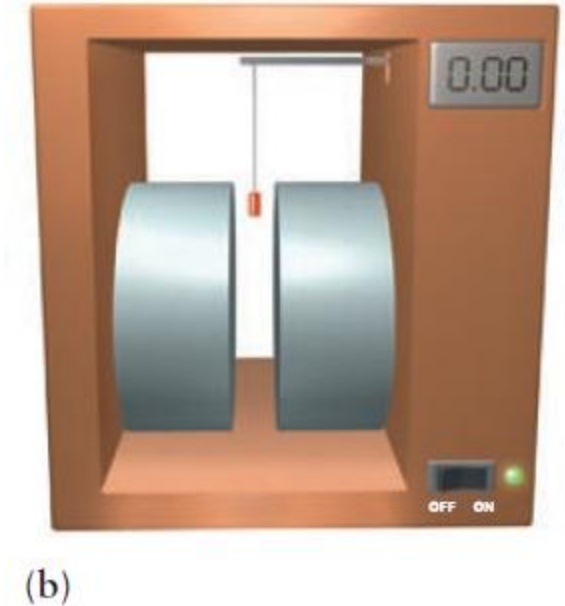
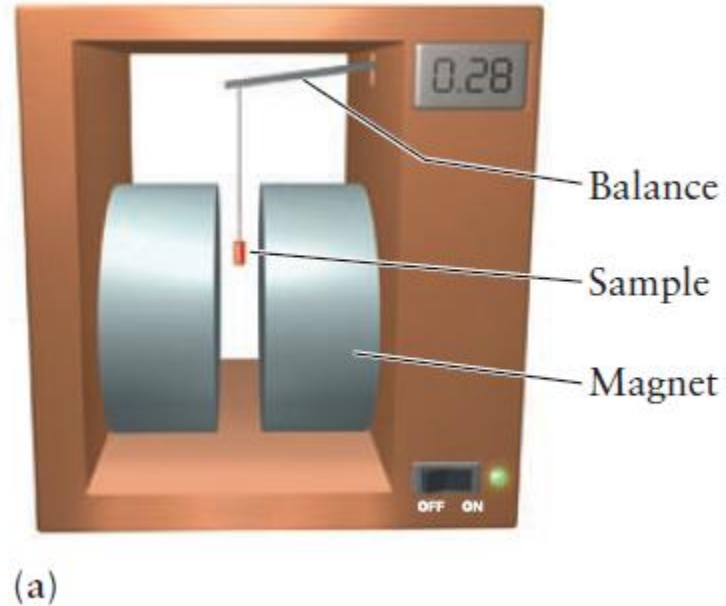
Square-planar: d^8 ions & strong field ligands.

- The low-spin configuration leaves the high-energy $d_{x^2-y^2}$ orbital vacant is the most stable.

The octahedral complex ions $[\text{FeCl}_6]^{3-}$ and $[\text{Fe}(\text{CN})_6]^{3-}$ are both paramagnetic, but the former is high spin and the latter is low spin. Identify the d -electron configurations in these two octahedral complex ions. In which is the octahedral field splitting greater? How does the CFSE differ between the complexes?

Crystal Field Theory: Magnetic Properties

FIGURE 8.22 Samples are initially inserted about halfway into a magnetic field to create a field gradient that exerts a force on magnetic dipoles. (a) Paramagnetic compounds are attracted by the field and “weigh” more than they do with no field present. (b) Diamagnetic compounds are repelled slightly by the field and “weigh” less than they do with no field present.



The measurements provide not only qualitative characterization of a sample's magnetic properties, but also a quantitative value for its **magnetic susceptibility 磁化率**

Major points of today's lecture

- **Bonding in transition metal compounds and coordination complexes**
 1. **Chemistry of the Transition Metals**
 2. **Introduction to Coordination Chemistry**
 3. **Structures of Coordination Complexes**
 4. **Crystal Field Theory: Optical and Magnetic Properties**
 5. **Optical Properties and the Spectrochemical Series**
 6. **Bonding in Coordination Complexes**

Optical Properties and the Spectrochemical Series



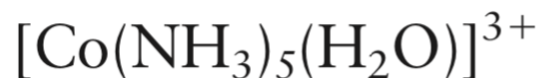
FIGURE 8.23 Several colored coordination compounds. (clockwise from top left) They are $\text{Cr}(\text{CO})_6$ (white), $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$ (green), $[\text{Co}(\text{en})_3]\text{I}_3$ (orange), $[\text{Co}(\text{NH}_3)_5(\text{OH}_2)]\text{Cl}_3$ (red), and $\text{K}_3[\text{Fe}(\text{CN})_6]$ (red-orange).



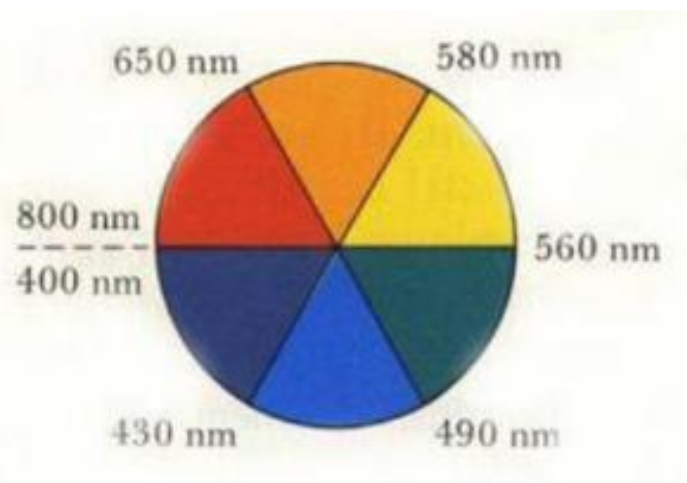
Orange



A green form and a violet form



Purple

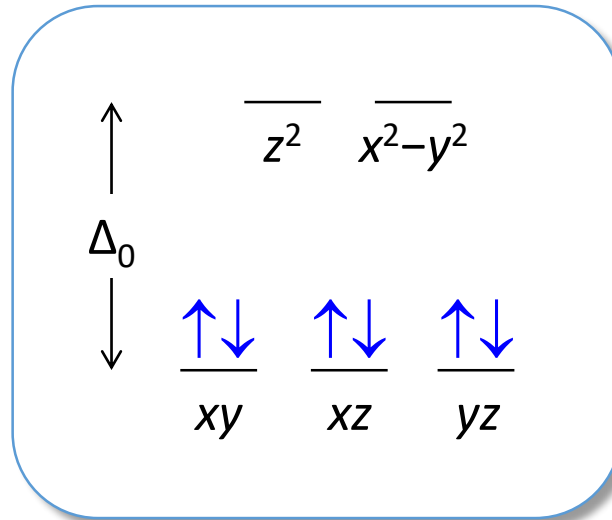
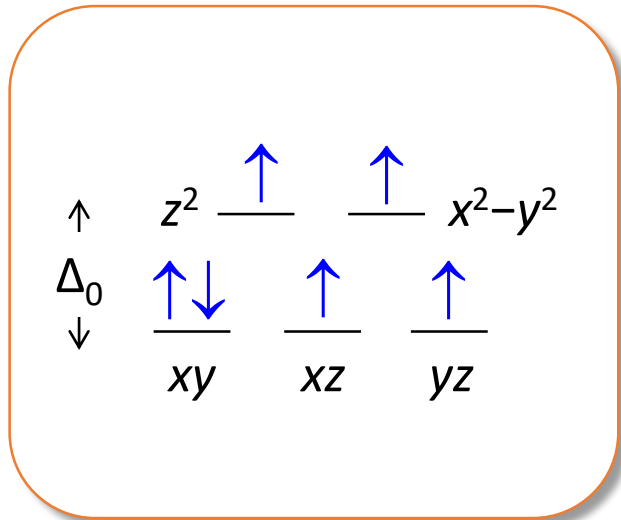


The colors depend on the

- **oxidation state of the metal ion**
- **number and nature of the ligands**
- **geometry of the complex**

Optical Properties and the Spectrochemical Series

The wavelength of the strongest absorption is called λ_{max} and it is related to Δ_o

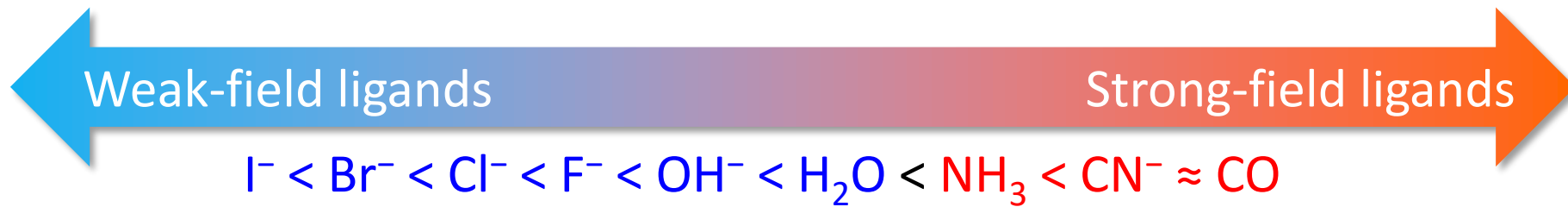


$$\Delta_o = h\nu = hc/\lambda_{\text{max}}$$

$[\text{Co}(\text{NH}_3)_6]^{3+}$	Orange
$[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$	A green form and a violet form
$[\text{Co}(\text{NH}_3)_5(\text{H}_2\text{O})]^{3+}$	Purple

High spin

Low spin

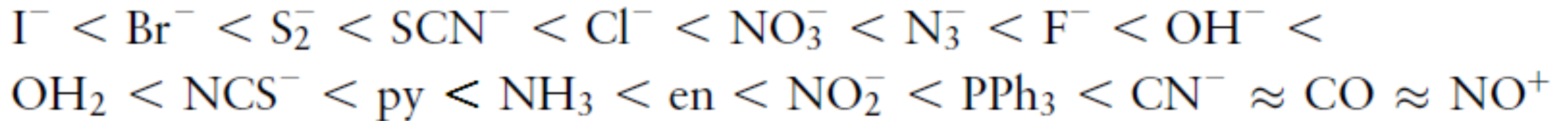


The smaller halide ions can approach the metal ion more closely: greater electrostatic repulsion.

Optical Properties and the Spectrochemical Series

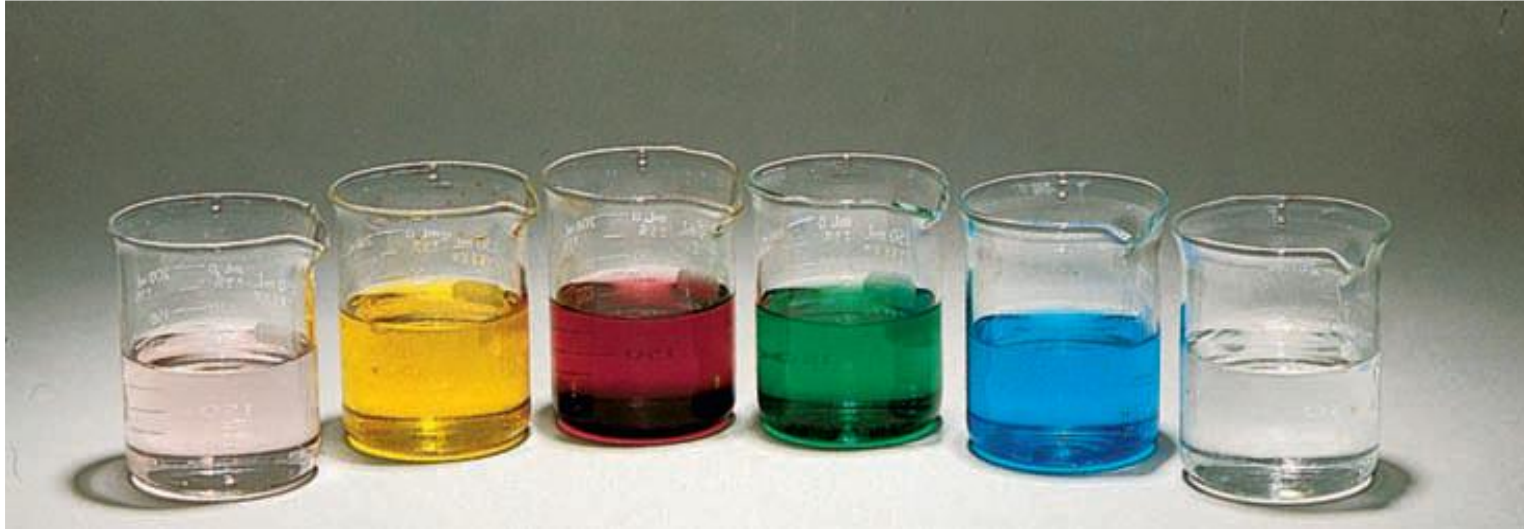
The strength of ligands was obtained by comparing the optical absorption spectra of $[\text{Co(III)(NH}_3)_5\text{X}]^{n+}$.

The strength of ligand X could be measured directly because all other interactions and the geometry of the complex remained constant.



Weak-field ligands (high-spin) Intermediate-field ligands Strong-field ligands (low-spin)

Optical Properties and the Spectrochemical Series



d¹⁰ complexes (Zn²⁺ or Cd²⁺) are colorless because all of the d levels (t_{2g} and e_g) are filled. There are no empty orbitals available to accept an excited electron, which means that the absorption is weak or nonexistent.

High-spin d⁵ complexes such as $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ also show weak absorption because promotion from a filled t_{2g} level to an empty e_g level would require a spin flip to satisfy the Pauli principle.

Predict which of the following octahedral complexes has the shortest λ_{max} : $[\text{FeF}_6]^{3-}$, $[\text{Fe}(\text{CN})_6]^{3-}$, $[\text{Fe}(\text{OH}_2)_6]^{3+}$.

Optical Properties and the Spectrochemical Series

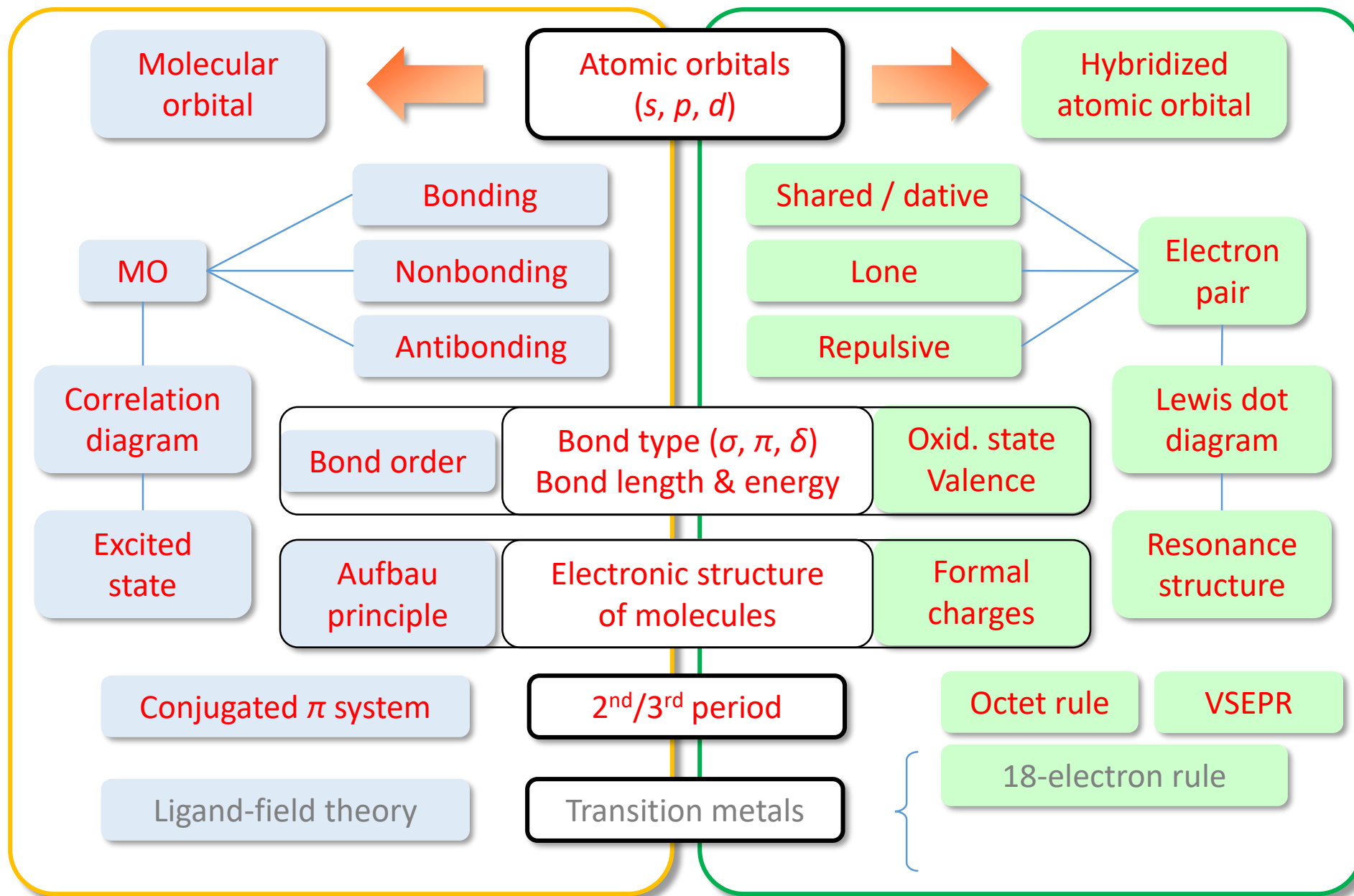
Crystal field theory was developed, in part, to explain the colors of transition metal complexes. Its failure to predict trends in the optical absorption of a series of related compounds stimulated the development of **ligand field** and **molecular orbital theories** and their application in coordination chemistry.

Major points of today's lecture

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Molecular Orbital (MO)

Valence Bond (VB) / Hybridization



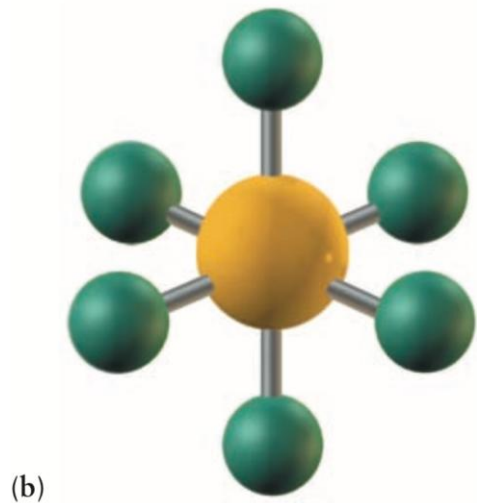
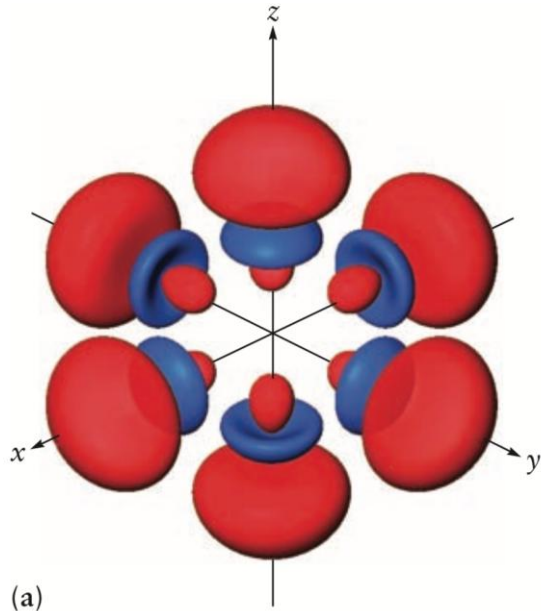
Bonding in Coordination Complexes: valence bond theory

Valence Bond Theory

- Coordination complexes: **a central atom surrounded by ligands** in a symmetric arrangement, forming **hybrid orbitals on the central atom with the appropriate symmetry to bond to these ligands**
- Often, **little interaction occurs among metal–ligand bonds**, so the **local description** is reasonable.

Bonding in Coordination Complexes: valence bond theory

d^2sp^3 hybrid orbitals



$$\chi_1 = \sqrt{\frac{1}{6}} [s + \sqrt{3} (p_z) + \sqrt{2} (d_{z^2})]$$

$$\chi_2 = \sqrt{\frac{1}{6}} \left[s + \sqrt{3} (p_z) - \sqrt{\frac{1}{2}} (d_{z^2}) + \sqrt{\frac{3}{2}} (d_{x^2-y^2}) \right]$$

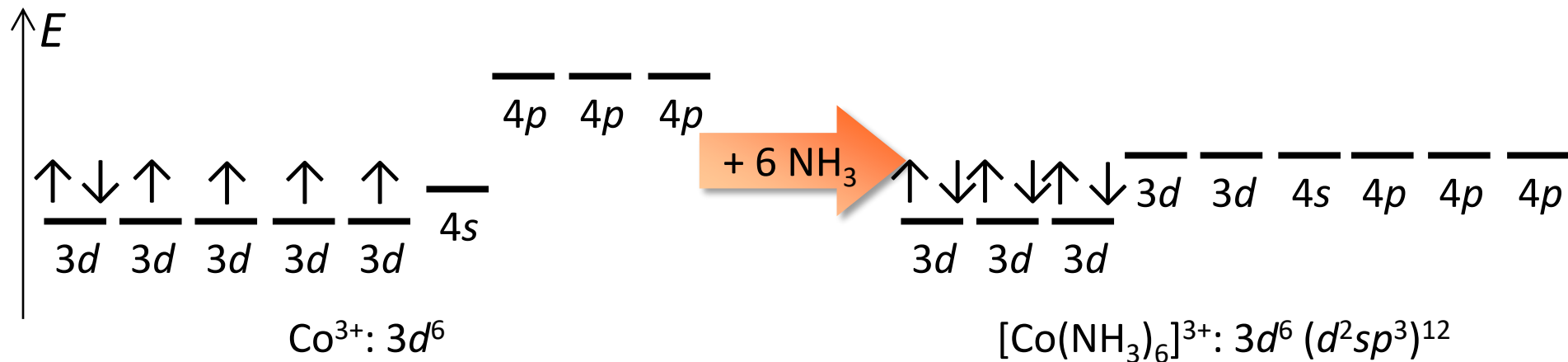
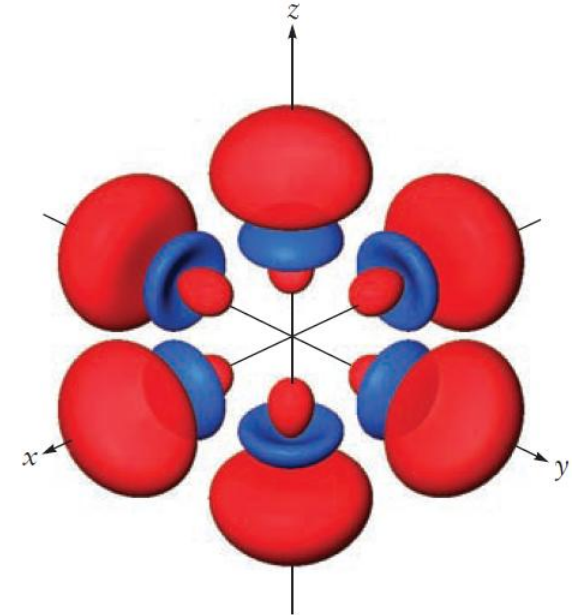
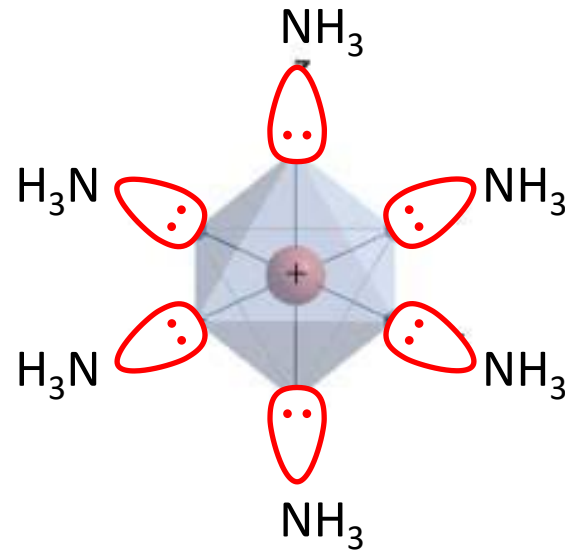
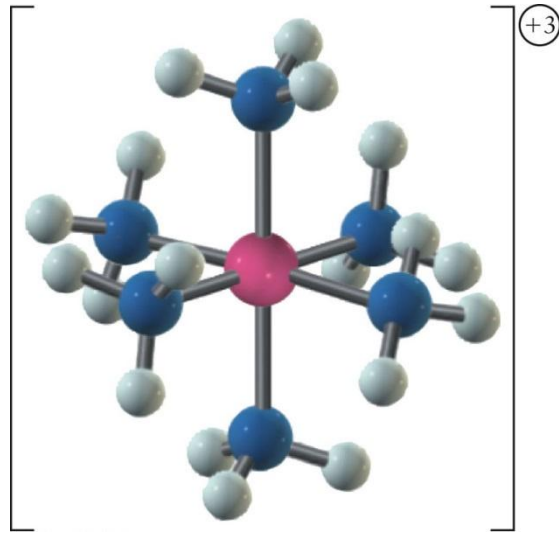
$$\chi_3 = \sqrt{\frac{1}{6}} \left[s + \sqrt{3} (p_z) - \sqrt{\frac{1}{2}} (d_{z^2}) - \sqrt{\frac{3}{2}} (d_{x^2-y^2}) \right]$$

$$\chi_4 = \sqrt{\frac{1}{6}} \left[s - \sqrt{3} (p_z) - \sqrt{\frac{1}{2}} (d_{z^2}) + \sqrt{\frac{3}{2}} (d_{x^2-y^2}) \right]$$

$$\chi_5 = \sqrt{\frac{1}{6}} \left[s - \sqrt{3} (p_z) - \sqrt{\frac{1}{2}} (d_{z^2}) - \sqrt{\frac{3}{2}} (d_{x^2-y^2}) \right]$$

$$\chi_6 = \sqrt{\frac{1}{6}} [s - \sqrt{3} (p_z) + \sqrt{2} (d_{z^2})]$$

Orbital Hybridization in $[\text{Co}(\text{NH}_3)_6]^{3+}$



Bonding in Coordination Complexes: valence bond theory

T A B L E 8.7

Examples of Hybrid Orbitals and Bonding in Complexes

Coordination Number	Hybrid Orbital	Configuration	Examples
2	sp	Linear	$[\text{Ag}(\text{NH}_3)_2]^+$
3	sp^2	Trigonal planar	BF_3 , NO_3^- , $[\text{Ag}(\text{PR}_3)_3]^+$
4	sp^3	Tetrahedral	$\text{Ni}(\text{CO})_4$, $[\text{MnO}_4]^-$, $[\text{Zn}(\text{NH}_3)_4]^{2+}$
4	dsp^2	Square planar	$[\text{Ni}(\text{CN})_4]^{2-}$, $[\text{Pt}(\text{NH}_3)_4]^{2+}$
5	dsp^3	Trigonal bipyramidal	TaF_5 , $[\text{CuCl}_5]^{3-}$, $[\text{Ni}(\text{PEt}_3)_2\text{Br}_3]$
6	d^2sp^3	Octahedral	$[\text{Co}(\text{NH}_3)_6]^{3+}$, $[\text{PtCl}_6]^{2-}$

From G.E. Kimball, Directed valence. *J. Chem. Phys.* 1940, 8, 188.

Homework: read “principle of modern chemistry”, chapter 8